

Algebraic Geometry

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Based on the book “The Rising Sea: Foundations of Algebraic Geometry” by Ravi Vakil

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1 Sheaves

1.1 Definitions

Definition 1.1.1

For a topological space X , its set of open subsets $\text{open}(X)$ is partially ordered by inclusion, and thus forms a small category. It is also a cofiltered category: for every $U, V \in \text{open}(X)$ they have a lower bound $U \cap V$.

Definition 1.1.2

Let X be a topological space, and $\mathcal{C}$ a category. A **$\mathcal{C}$ -valued presheaf on X** is a contravariant functor $\mathcal{F}: \text{open}(X)^{\text{op}} \rightarrow \mathcal{C}$. It is customary to denote $\mathcal{F}(U \subseteq V): \mathcal{F}(V) \rightarrow \mathcal{F}(U)$ by res_U^V . When $\mathcal{C}$ is a concrete category (i.e. its objects are sets), elements of $\mathcal{F}(U)$ are called **sections of $\mathcal{F}$ over U** , and for a section $s \in \mathcal{F}(V)$ we write $s|_U$ for $\text{res}_U^V(s)$.

A **morphism of presheaves** is a natural transformation.

Denote the category of $\mathcal{C}$ -valued presheaves on X by $\text{Psh}_{\mathcal{C}}(X)$.

Example 1.1.3

Let E be a set (or a topological space, or any mathematical object which lies in the same category as X and its open subsets). We can then define the presheaf of functions on E by $\mathcal{F}(U) = \text{hom}(U, E)$: the set of all functions $U \rightarrow E$. The restriction maps are very literally restriction: for $U \subseteq V$ and $f: V \rightarrow E$, we restrict f to U to obtain a function $f|_U: U \rightarrow E$.

Definition 1.1.4

Let $\mathcal{F}$ be a presheaf over X valued in a category $\mathcal{C}$ with filtered colimits, and $p \in X$. Then $\mathcal{U}(p) = \{U \subseteq \text{open} \mid p \in U\}$, the set of neighborhoods of p , is a cofiltered subcategory of $\text{open}(X)$, and so $\mathcal{F}$ restricts to a functor from $\mathcal{U}(p)^{\text{op}}$. Define $\mathcal{F}$'s **stalk** at p to be the filtered colimit

$$\mathcal{F}_p = \text{colim}_{U \in \mathcal{U}(p)} \mathcal{F}(U) \in \mathcal{C}.$$

For concrete categories, this is the set of **germs** of $\mathcal{F}$ at p :

$$\mathcal{F}_p = \{(U, s) \mid U \in \mathcal{U}(p), s \in \mathcal{F}(U)\},$$

modulo the relation where we identify (s, U) and (t, V) if there is a subset $p \in W \subseteq U, V$ such that $s|_W = t|_W$.

Taking colimits is functorial, so the correspondence $p \mapsto \mathcal{F}_p$ is functorial. Explicitly $\varphi: \mathcal{F} \rightarrow \mathcal{G}$ is a morphism of sheaves over X valued in a concrete category, and $p \in X$ we define $\varphi_p: \mathcal{F}_p \rightarrow \mathcal{G}_p$ by

$$\varphi_p(U, s) = (U, \varphi_U(s)).$$

Definition 1.1.5

A presheaf $\mathcal{F}$ on X (valued in a concrete category) is a **sheaf** if it satisfies the **sheaf axiom**: for every

open subset $U \subseteq X$ and open cover $\{U_i\}$ of U and sections $s_i \in \mathcal{F}(U_i)$, if $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ for all i, j then there exists a *unique* $s \in \mathcal{F}(U)$ such that $s|_{U_i} = s_i$ for all i .

We can split the sheaf axiom in two:

- (1) the **identity axiom**: if $\{U_i\}$ is an open cover of U , and $s_1, s_2 \in \mathcal{F}(U)$ are such that $s_1|_{U_i} = s_2|_{U_i}$ for all i , then $s_1 = s_2$.
- (2) the **gluability axiom**: the sheaf axiom without requiring uniqueness.

We can also state the sheaf axiom categorically. For every open cover $\{U_i\}$ of U , we can consider the following diagram:

$$\mathcal{F}(U) \xrightarrow{\text{res}} \prod_i \mathcal{F}(U_i) \xrightleftharpoons[\text{res}_j]{\text{res}_i} \prod_{i,j} \mathcal{F}(U_i \cap U_j),$$

where $\mathcal{F}(U) \xrightarrow{\text{res}} \prod_i \mathcal{F}(U_i)$ is the map $s \mapsto (s|_{U_i})_i$, res_1 and res_2 are the maps

$$\text{res}_1(s_i)_i = (s_i|_{U_i \cap U_j})_{i,j}, \quad \text{res}_2(s_i)_i = (s_j|_{U_i \cap U_j})_{i,j}.$$

Categorically, res_1 's coordinate to $\mathcal{F}(U_i \cap U_j)$ composes the i th projection with $\text{res}_{U_i \cap U_j}^{U_i}$, and res_2 's coordinate composes the j th projection with $\text{res}_{U_i \cap U_j}^{U_j}$. These maps are defined for any presheaf, and we can see that $\mathcal{F}$ is a presheaf iff this is an equalizer diagram.

Indeed, the equalizer of $\text{res}_1, \text{res}_2$ are all $(s_i)_i$ such that $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$, i.e. a family of sections which agree on the overlaps. And there is an $s \in \mathcal{F}(U)$ whose projections $s|_{U_i}$ are s_i when the gluing axiom is satisfied, and it is unique (so an embedding, and thus an equalizer) when the identity lemma is satisfied.

This allows us to generalize sheaves to any category $\mathcal{C}$ with arbitrary products.

Remark 1.1.6

Alternatively for an open cover $\mathcal{U} = \{U_i\}_{i \in I}$ of U , we can define a subcategory of $\text{open}(X)$ consisting of objects $U_i, U_i \cap U_j$ with morphisms $U_i \cap U_j \subseteq U_i$. Then $\mathcal{F}$ restricts to a functor $\mathcal{F}|_{\mathcal{U}}: \mathcal{U}^{\text{op}} \rightarrow \mathcal{C}$. Then the sheaf condition says that the cone $\mathcal{F}(U) \Rightarrow \mathcal{F}|_{\mathcal{U}}$ whose components are the restriction maps res , is a limiting cone.

The category of $\mathcal{C}$ -valued sheaves on X is denoted $\text{Sh}_{\mathcal{C}}(X)$, it is a full subcategory of $\text{Psh}_{\mathcal{C}}(X)$; meaning sheaf morphisms are just presheaf morphisms between sheaves. The stalk of a sheaf at a point is just the stalk of it as a presheaf.

Example 1.1.7

If $\mathcal{F}$ is a presheaf on X , and $U \subseteq X$ is open, we define $\mathcal{F}$'s **restriction** to U to be the presheaf $\mathcal{F}|_U$ on U defined for $V \subseteq U$ by $\mathcal{F}|_U(V) = \mathcal{F}(V)$ (it is just the restriction of the functor $\mathcal{F}$ to the subcategory $\text{open}(U)$). If $\mathcal{F}$ is a sheaf, its restriction is clearly a sheaf as well.

Example 1.1.8

Let X be a topological space, $p \in X$, and S a set (or a group, module, etc.). We define the **skyscraper sheaf** p_*S by

$$p_*S(U) = \begin{cases} S & \text{if } p \in U \\ * & \text{else} \end{cases},$$

where $*$ is a single-point set; the restrictions are given by the identities or the unique morphism $S \rightarrow *$. This is indeed a sheaf: if $p \in U$ and $\{U_i\}$ is a covering of U with $s_i \in \mathcal{F}(U_i)$ such that $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$

then for every $p \in U_i$ we have a single value $s \in S$ which is equal to all $s_i \in \mathcal{F}(U_i)$, this is the glued section. Otherwise $p \notin U$ and s_i are all the unique element in $*$, so s must be too.

Example 1.1.9

Let X be a topological space and S a set. The **constant presheaf** associated with S is the presheaf $\underline{S}_{\text{pre}}$ which maps all open sets U to S . This is a presheaf, but not a sheaf in general. For example if $X = \{p, q\}$ is the two point space with the discrete topology then take $s_p \neq s_q$; they cannot be glued to a global section on X .

If we instead define $\underline{S}(U)$ to be the set of locally constant functions $U \rightarrow S$, then this is a sheaf, called the **constant sheaf**. It is just the sheaf of continuous functions $\underline{S}(U) = C(U, S)$ with S given the discrete topology. The sheaf of continuous functions $C(\bullet, Y)$ is a sheaf in general (by the gluing lemma from topology).

Example 1.1.10

Let $\pi: X \rightarrow Y$ be a continuous map, and $\mathcal{F}$ a presheaf over X . Define the **pushforward** (or **direct image**) presheaf $\pi_*\mathcal{F}$ by

$$\pi_*\mathcal{F}(V) = \mathcal{F}(\pi^{-1}V),$$

for $V \subseteq Y$, and $\text{res}_W^V = \text{res}_{\pi^{-1}W}^{\pi^{-1}V}$. This is clearly a presheaf; if $\mathcal{F}$ is a sheaf it is also a sheaf. Indeed, if $V \subseteq Y$ is open and $\{V_i\}_i$ is a cover with $s_i \in \pi_*\mathcal{F}(V_i) = \mathcal{F}(\pi^{-1}V_i)$. Then the glue of these is unique, because $\mathcal{F}$ is a sheaf.

The skyscraper sheaf p_*S is the pushforward of the constant sheaf $\underline{S}$ by the inclusion $\{p\} \rightarrow X$.

An important property of the direct image sheaf is that it is functorial: if $\varphi: \mathcal{F} \rightarrow \mathcal{G}$ is a morphism of presheaves over X , then $\pi_*\varphi: \pi_*\mathcal{F} \rightarrow \pi_*\mathcal{G}$ defined by $\pi_*\varphi_V = \varphi_{\pi^{-1}V}$ is a morphism of presheaves.

Note that if $\pi: X \rightarrow Y$ is continuous, $\mathcal{F}$ is a sheaf over X , and $\pi(p) = q$, then there is a natural morphism of stalks

$$(\pi_*\mathcal{F})_q \rightarrow \mathcal{F}_p.$$

Indeed, the left-hand side is

$$\text{colim}_{q \in V} \mathcal{F}(\pi^{-1}V) = \text{colim}_{p \in \pi^{-1}V} \mathcal{F}(\pi^{-1}V).$$

Now a cone under $\mathcal{F}$ defines a unique cone under $\pi_*\mathcal{F}$ simply by restricting, and since $\text{colim} \pi_*\mathcal{F}$ is the initial cone under $\pi_*\mathcal{F}$ there is a canonical map to this restricted cone, in particular for $\text{colim} \mathcal{F}$.

In more concrete terms, $(\pi_*\mathcal{F})_q$ has elements (s, V) for $s \in \mathcal{F}(\pi^{-1}V)$, which we then map to $(s, \pi^{-1}V) \in \mathcal{F}_p$.

Definition 1.1.11

If $\mathcal{O}_X$ is a sheaf of rings over a topological space X (i.e. a sheaf over X valued in the category of rings), then the pair $(X, \mathcal{O}_X)$ is called a **ringed space**; $\mathcal{O}_X$ is called the **structure sheaf**. Sections of $\mathcal{O}_X$ over $U \subseteq X$ are called **functions** on U , global sections are called **global functions**.

Definition 1.1.12

If $\mathcal{O}_X$ is a ringed space, an $\mathcal{O}_X$ -**module** is a sheaf of Abelian groups $\mathcal{F}$ over X such that for each $U \subseteq X$, $\mathcal{F}(U)$ is an $\mathcal{O}_X(U)$ -module, and all of these modules are well-behaved in that when $U \subseteq V$, the following diagram commutes:

$$\begin{array}{ccc}
\mathcal{O}_X(V) \times \mathcal{F}(V) & \xrightarrow{\text{mult}} & \mathcal{F}(V) \\
\text{res}_{V,U} \times \text{res}_{V,U} \downarrow & & \downarrow \text{res}_{V,U} \\
\mathcal{O}_X(U) \times \mathcal{F}(U) & \xrightarrow{\text{mult}} & \mathcal{F}(U)
\end{array}$$

The horizontal arrows are $\mathcal{O}_X(U)$'s action on $\mathcal{F}(U)$.

The commutative diagram means that

$$(s \cdot x)|_U = (s|_U) \cdot (x|_U),$$

for $s \in \mathcal{O}_X(V)$, $x \in \mathcal{F}(V)$.

Note that just as modules generalize the concept of an Abelian group (Abelian groups are $\mathbb{Z}$ -modules), $\mathcal{O}_X$ -modules generalize the concept of a sheaf of Abelian groups. Indeed, a sheaf of Abelian groups is just an $\mathbb{Z}$ -module. Every $\mathbb{Z}$ -module is a sheaf of Abelian groups by definition, and conversely a sheaf of Abelian groups has a unique $\mathbb{Z}$ -module structure where for $U \subseteq X$ connected we have $\mathbb{Z}(U) \cong \mathbb{Z}$ which has a unique multiplication on $\mathcal{F}(U) \in \mathbf{Ab}$. We extend this to arbitrary $U \subseteq X$ by gluing.

Proposition 1.1.13

If $\mathcal{F}$ is an $\mathcal{O}_X$ -module, then for each $p \in X$, $\mathcal{F}_p$ is an $\mathcal{O}_{X,p}$ -module.

Proof

The condition that $\mathcal{F}$ be an $\mathcal{O}_X$ -module means that multiplication is a natural map $\mathcal{O}_X \times \mathcal{F} \rightarrow \mathcal{F}$. Colimits are taken pointwise in functor categories (in particular for the product of functors), so this gives a natural map $\mathcal{O}_{X,p} \times \mathcal{F}_p \rightarrow \mathcal{F}_p$. The natural map is given by mapping $(s, U), (x, V)$ to $(sx, U \cap V)$, which is easily seen to give a module structure. ■

Example 1.1.14

Consider a smooth manifold X ; this defines a ringed space $(X, \mathcal{O}_X)$ where $\mathcal{O}_X$ is the sheaf of smooth real-valued functions: $\mathcal{O}_X(U) = C^\infty(U)$. These are all real algebras, so we could view this as an $\mathbb{R}$ -ringed space.

If $\pi: V \rightarrow X$ is a vector bundle over X , then consider the sheaf of smooth local sections of π :

$$\Gamma(U) = \{\sigma: U \rightarrow V \text{ smooth} \mid \pi \circ \sigma = \text{id}_U\}.$$

Then Γ is an $\mathcal{O}_X$ -module: it is a classical result that a local section defined on U can be multiplied by a smooth real-valued function defined on U .

1.2 The category of sheaves

As discussed, presheaves form a category for which sheaves form a full subcategory. We will denote the category of $\mathcal{C}$ -valued presheaves over a space X by $\mathcal{C}_X^{\text{pre}}$ (instead of $\mathbf{Psh}_{\mathcal{C}}(X)$), and we will denote the full subcategory of sheaves by $\mathcal{C}_X \subseteq \mathcal{C}_X^{\text{pre}}$.

Note that if $\mathcal{F}, \mathcal{G} \in \mathcal{C}_X$ are sheaves (or presheaves), then the hom-set of sheaf morphisms $\mathcal{F} \rightarrow \mathcal{G}$ is indeed a set, since X as a category is small (functor categories under small categories are locally small). Furthermore, if $\mathcal{C}$ is enriched over some monoidal category (in our case, usually $\mathbf{Ab}$) so is $\mathcal{C}_X$. This is because it is true in general for functor categories over enriched categories.

Definition 1.2.1

Let $\mathcal{F}, \mathcal{G}$ be $\mathcal{C}$ -valued sheaves defined on X . Define their **sheaf-hom** to be the sheaf whose sections are sheaf morphisms,

$$\mathbf{Hom}(\mathcal{F}, \mathcal{G})(U) = \text{hom}_{\mathcal{C}_X}(\mathcal{F}|_U, \mathcal{G}|_U).$$

As discussed, if $\mathcal{C}$ is enriched over a monoidal category $\mathcal{V}$ then so is $\mathcal{C}_X$, and so $\mathbf{Hom}(\mathcal{F}, \mathcal{G})$ takes values in $\mathcal{V}$. In particular, if $\mathcal{F}, \mathcal{G}$ are sheaves of Abelian groups, then so is their sheaf-hom.

Proposition 1.2.2

The sheaf-hom is indeed a sheaf.

Proof

We first show it is a sheaf: if $U \subseteq V$ then a sheaf morphism $\mathcal{F}|_V \rightarrow \mathcal{G}|_V$ restricts to a sheaf morphism $\mathcal{F}|_U \rightarrow \mathcal{G}|_U$ in the obvious way: its components are the same but only taken on subsets of U . This is clearly a presheaf.

To show that it is a sheaf, let $\{U_i\}$ be an open cover for $U \subseteq X$, and $\alpha_i: \mathcal{F}|_{U_i} \rightarrow \mathcal{G}|_{U_i}$ be sheaf morphisms such that $\alpha_i|_{U_i \cap U_j} = \alpha_j|_{U_i \cap U_j}$ for all i, j . In order to glue them together for a sheaf morphism $\alpha: \mathcal{F}_U \rightarrow \mathcal{G}_U$ we need to define α_W for all $W \subseteq U$. For $s \in \mathcal{F}(W)$ consider $s_i = s|_{U_i} \in \mathcal{F}(U_i \cap W)$, and its image under α_i to $\mathcal{G}(U_i \cap W)$, call it t_i . Then since $U_i \cap W$ cover W and since α is a morphism as well as $\mathcal{G}$ being a sheaf, we can uniquely glue t_i together to get a $t \in \mathcal{G}(W)$; this will be the image of s .

It is not hard to see that this is the unique gluing of α_i , and is itself a sheaf morphism. ■

Note:

In the above proof we needed only for $\mathcal{G}$ to be a sheaf, $\mathcal{F}$ can be a presheaf.

Furthermore **Hom** forms a bifunctor

$$\mathbf{Hom}: \mathcal{C}_X^{\text{op}} \times \mathcal{C}_X \rightarrow \mathbf{Set}_X,$$

or V_X instead of $\mathbf{Set}_X$ when $\mathcal{C}$ is enriched over V .

Note that if $\mathcal{O}_X$ is a ringed space, it is also a module over itself. Thus if $\mathcal{F}$ is an $\mathcal{O}_X$ module, we can form its **dual**:

$$\mathcal{F}^\vee = \mathbf{Hom}_{\text{Mod}_{\mathcal{O}_X}}(\mathcal{F}, \mathcal{O}_X).$$

We now focus on sheaves of Abelian groups. It is a classical result that functor categories over Abelian categories are Abelian, thus $\mathbf{Ab}_X^{\text{pre}}$ is an Abelian category. That is to say, the hom-sets (not hom-sheaves) have an Abelian group structure: for $\varphi, \psi: \mathcal{F} \rightarrow \mathcal{G}$ we can define $\varphi + \psi$ and $-\varphi$ by $(\varphi + \psi)(U) = \varphi(U) + \psi(U)$ and $(-\varphi)(U) = -\varphi(U)$. And we can form kernels and cokernels so that all monomorphisms are kernels and all epimorphisms are cokernels.

As with all limits in functor categories, these are done coordinate-wise. That is to say, if we denote by $\ker_{\text{pre}}$ and $\text{coker}_{\text{pre}}$ the kernel and cokernel of Abelian presheaf morphisms, we have

$$(\ker_{\text{pre}} \varphi)(U) = \ker(\varphi(U)), \quad (\text{coker}_{\text{pre}} \varphi)(U) = \text{coker}(\varphi(U)).$$

Remark 1.2.3

Let I be an indexing category and $\mathcal{C}$ be a category with arbitrary pullbacks. It is easy to see that a $\mathcal{C}$ -morphism $f: X \rightarrow Y$ is monic iff the following square is a pullback:

$$\begin{array}{ccc}
X & \xrightarrow{\quad \text{id} \quad} & X \\
\text{id} \downarrow & & \downarrow f \\
X & \xrightarrow{\quad f \quad} & Y
\end{array}$$

Because the functor category $[I, \mathcal{C}]$ also has arbitrary pullbacks, a natural transformation $\alpha: \mathcal{F} \Rightarrow \mathcal{G}$ is monic iff a similar diagram is a pullback. Because limits in a functor category are taken coordinate-wise, this is iff $\alpha(i)$ is monic for all $i \in I$.

Thus a natural transformation is monic iff its coordinates are.

Similarly, if $\mathcal{C}$ has arbitrary pushouts, natural transformations in $[I, \mathcal{C}]$ are epic iff their coordinates are.

In lieu of this note, we see that $\mathbf{Ab}_X^{\text{pre}}$ forms an Abelian category. Indeed, if $\varphi: \mathcal{F} \Rightarrow \mathcal{G}$ is a monic morphism of presheaves, then $\varphi(U): \mathcal{F}(U) \Rightarrow \mathcal{G}(U)$ is a monic morphism of Abelian groups for all $U \subseteq X$. Thus $\varphi(U) = \ker \text{coker } \varphi(U)$ for all $U \subseteq X$ and thus $\varphi = \ker \text{coker } \varphi$ as desired. Similar for epic morphisms.

Many properties of $\mathbf{Ab}_X^{\text{pre}}$ are seen to be pointwise. For one, if we set an open set $U \subseteq X$, the functor $\mathbf{Ab}_X^{\text{pre}} \rightarrow \mathbf{Ab}$ mapping $\mathcal{F} \mapsto \mathcal{F}(U)$ is an exact functor. Indeed, it is additive and preserves kernels and cokernels because these are taken pointwise.

Furthermore, a sequence of presheaves

$$\mathcal{F}_1 \rightarrow \mathcal{F}_2 \rightarrow \cdots \rightarrow \mathcal{F}_n$$

is exact iff $\mathcal{F}_1(U) \rightarrow \cdots \rightarrow \mathcal{F}_n(U)$ is for all $U \subseteq X$. Again, this is an immediate consequence of kernels and cokernels (and thus images) being taken coordinate-wise.

Everything we have said holds for presheaves valued in any Abelian category $\mathcal{A}$ (recall that Abelian categories have all finite limits and colimits). But we are more interested in the category of sheaves, in which case this is less obvious.

Note that the constant presheaf $\mathbf{0}_{\text{pre}}$ is a sheaf, so we can denote it by $\mathbf{0}$ without confusion. In order to show that $\mathcal{A}_X$ is an Abelian category, we must first show that kernels and cokernels exist in it. The following proposition shows that the kernel of a sheaf morphism is the same as the kernel of a presheaf morphism, and thus we get the existence of kernels for free.

Proposition 1.2.4

If $\varphi: \mathcal{F} \rightarrow \mathcal{G}$ is a morphism of sheaves, then $\ker_{\text{pre}} \varphi$ is a sheaf. Thus $\ker \varphi = \ker_{\text{pre}} \varphi$ forms a kernel in the category of sheaves.

Proof

We show this for $\mathcal{A} = \mathbf{Ab}$: if $\{U_i\}_i$ is an open cover of $U \subseteq X$ and $s_i \in \ker \varphi(U_i)$ are such that $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ for all i, j , then by gluing they form a section $s \in \mathcal{F}(U)$. And by naturality

$$\varphi(U)(s)|_{U_i} = \varphi(U_i)(s|_{U_i}) = \varphi(U_i)(s_i) = 0.$$

The unique gluing of $0 \in \mathcal{G}(U_i)$ is $0 \in \mathcal{G}(U)$, so that $\varphi(U)(s) = 0$, meaning $s \in \ker \varphi(U)$ as desired. $\blacksquare$

Remark 1.2.5

A generalization of the above proposition shows that limits taken in the category of presheaves of sheaves forms a sheaf. That is, if $\Phi: I \rightarrow \mathcal{C}_X$ is a functor, then consider the inclusion $\iota: \mathcal{C}_X \hookrightarrow \mathcal{C}_X^{\text{pre}}$. Then $\lim \iota \circ \Phi$ is a sheaf, so $\lim \Phi$ exists in $\mathcal{C}_X$.

Thus if $\mathcal{C}$ has arbitrary pullbacks, so too does $\mathcal{C}_X^{\text{pre}}$ and thus so too does $\mathcal{C}_X$. It then follows that a morphism

of sheaves is monic iff its monic as a morphism of presheaves (considering the pullback condition remark 1.2.3) iff each component is monic (again by remark 1.2.3).

Unfortunately we now run in to our first hurdle: presheaf cokernels are not sheaves.

Example 1.2.6

Consider $\mathbb{C}$ with its usual topology, and let $\mathcal{O}_{\mathbb{C}}$ be the sheaf of holomorphic functions. Let $\mathcal{F}$ be the presheaf of functions admitting a holomorphic logarithm. Consider the following exact sequence of presheaves

$$0 \rightarrow \mathbb{Z} \hookrightarrow \mathcal{O}_{\mathbb{C}} \xrightarrow{\exp(2\pi i)} \mathcal{F} \rightarrow 0,$$

where the second map is $f \mapsto \exp(2\pi i f)$. This is indeed an exact sequence of presheaves: the first morphism is an embedding, and the second is epic (if $f = \exp(g)$ then $g/2\pi i \in \mathcal{O}_{\mathbb{C}}$ is an origin for f). The composition is zero, as $\exp(2\pi i n) = 0$, and if $\exp(2\pi i f) = 0$ then $f(x) \in \mathbb{Z}$ for all $x \in \mathbb{C}$. Because $\mathbb{Z}$ is discrete, this means $f \in \mathbb{Z}$ as desired.

This shows that $\mathcal{F}$ is the cokernel of $\mathbb{Z} \hookrightarrow \mathcal{O}_{\mathbb{C}}$: but it is not a sheaf! Indeed $\text{id}: \mathbb{C} \rightarrow 0 \rightarrow \mathbb{C}$ has a local logarithm at every point in its domain, but no global logarithm. Thus $\mathcal{F}$ fails the gluing axiom.

Thus for now we must wait to demonstrate that $\mathcal{A}_X$ is an Abelian category.

1.3 Sheafification

There is a natural map

$$\mathcal{F}(U) \rightarrow \mathcal{F}_p,$$

for every $p \in U$ given by mapping $s \in \mathcal{F}(U)$ to $(U, s) \in \mathcal{F}_p$. Together these assimilate into a map

$$\mathcal{F}(U) \rightarrow \prod_{p \in U} \mathcal{F}_p.$$

Proposition 1.3.1

The natural map above is injective when $\mathcal{F}$ is a sheaf.

Proof

Suppose that $s, t \in \mathcal{F}(U)$ such that $(U, s) = (U, t)$ in $\mathcal{F}_p$. Then there exists a neighborhood $p \in W_p \subseteq U$ such that $s|_{W_p} = t|_{W_p}$. If this is true for all p , then W_p form an open cover of U and by the identity axiom, $s = t$. ■

Definition 1.3.2

An element $(s_p)_p \in \prod_{p \in U} \mathcal{F}_p$ consists of **compatible germs** if for all $p \in U$, s_p has a representative $\tilde{s}_p \in \mathcal{F}(U_p)$ for $p \in U_p \subseteq U$ such that the germ of $\tilde{s}_p$ at $q \in U_p$ is s_q for all $q \in U_p$.

Equivalently, this says that there is an open cover $\{U_i\}$ of U and sections $f_i \in \mathcal{F}(U_i)$ such that if $p \in U_i$ then s_p is the germ of f_i at p . If s is a section of $\mathcal{F}$ over U , then $(s)_p$ consists of compatible germs.

Proposition 1.3.3

The natural map $\mathcal{F}(U) \rightarrow \prod_{p \in U} \mathcal{F}_p$'s image is precisely the compatible germs.

Proof

We already showed that every tuple in its image consists of compatible germs, now we show the converse. Suppose $(s_p)_p$ consists of compatible germs, so let $\tilde{s}_p \in \mathcal{F}(U_p)$ be a representation as in the definition. Then U_p cover U , and $\tilde{s}_p|_{U_p \cap U_q}$'s germ at q is s_q so that $\tilde{s}_p, \tilde{s}_q$ agree on a neighborhood of q and by symmetry in a neighborhood of p and q . ■

Proposition 1.3.4

Suppose $\mathcal{F}$ is a presheaf of sets and $\mathcal{G}$ a sheaf of sets, and $\varphi_1, \varphi_2: \mathcal{F} \rightarrow \mathcal{G}$ are two morphisms. Further suppose that φ_1, φ_2 induce the same map on each stalk; show that $\varphi_1 = \varphi_2$.

Proof

Consider the following diagram for $U \subseteq X$ open,

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\varphi_i} & \mathcal{G}(U) \\ \downarrow & & \downarrow \iota \\ \prod_{p \in U} \mathcal{F}_p & \xrightarrow{\prod \varphi_{i,p}} & \prod_{p \in U} \mathcal{G}_p \end{array}$$

for $i = 1, 2$. It is easily seen that this diagram commutes, and because $\varphi_{1,p} = \varphi_{2,p}$ for all $p \in U$ it follows that

$$\iota \circ \varphi_1 = \iota \circ \varphi_2$$

and since ι is injective by proposition 1.3.1, we get that $\varphi_1 = \varphi_2$ as desired. ■

Proposition 1.3.5

A morphism of sheaves of sets is injective iff it induces an injection on all stalks. Similarly it is an isomorphism iff it induces an isomorphism on all stalks.

Proof

If $\varphi: \mathcal{F} \rightarrow \mathcal{G}$ is an isomorphism, then by functoriality $\varphi_p: \mathcal{F}_p \rightarrow \mathcal{G}_p$ is as well for all p . Similarly if it is an injection then $\varphi_p(s_p) = \varphi_p(t_p)$ then $\varphi(s)$ and $\varphi(t)$ are equal in a subset $V \subseteq U$ so that $\varphi(s|_V) = \varphi(t|_V)$ and by injectivity this means that $s|_V = t|_V$ and thus $s_p = t_p$.

Conversely consider the commutative diagram

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\varphi} & \mathcal{G}(U) \\ \downarrow \iota_{\mathcal{F}} & & \downarrow \iota_{\mathcal{G}} \\ \prod_{p \in U} \mathcal{F}_p & \xrightarrow{\sim} & \prod_{p \in U} \mathcal{G}_p \end{array}$$

where the bottom horizontal arrow is the injection/isomorphism $\prod_{p \in U} \varphi_p$. Then $\iota_{\mathcal{G}} \circ \varphi$ is injective, and so φ is.

Now to show that φ is surjective, let $s \in \mathcal{G}(U)$. Then $\iota_{\mathcal{G}}(s) \in \prod_U \mathcal{G}_p$ consists of compatible germs, and mapping it back to $\prod_U \mathcal{F}_p$ we see that it still consists of compatible germs. Thus it lies in the image of $\iota_{\mathcal{F}}$ by proposition 1.3.3, and thus has a preimage in $\mathcal{F}(U)$. Because all of the maps are injective, this is the preimage of s in $\mathcal{F}(U)$ as desired. ■

Note:

It is not true that sheaves with isomorphic stalks are isomorphic: there may not be an isomorphism between the sheaves which induces these isomorphisms.

Theorem 1.3.6 (Sheafification)

The inclusion $\text{Set}_X \hookrightarrow \text{Set}_X^{\text{pre}}$ has a left adjoint, called **sheafification**: $\text{sh}: \text{Set}_X^{\text{pre}} \rightarrow \text{Set}_X$.

Proof

We need to show for any presheaf $\mathcal{F}$ there exists a sheaf $\mathcal{F}^{\text{sh}}$ and morphism $\text{sh}: \mathcal{F} \rightarrow \mathcal{F}^{\text{sh}}$ such that for every sheaf $\mathcal{G}$ and morphism $g: \mathcal{F} \rightarrow \mathcal{G}$ there exists a unique morphism $f: \mathcal{F}^{\text{sh}} \rightarrow \mathcal{G}$ such that the below diagram commutes:

$$\begin{array}{ccc} \mathcal{F} & \xrightarrow{\text{sh}} & \mathcal{F}^{\text{sh}} \\ & \searrow g & \downarrow \exists! f \\ & & \mathcal{G} \end{array}$$

We construct $\mathcal{F}^{\text{sh}}$ explicitly. For $U \subseteq X$ open, define

$$\mathcal{F}^{\text{sh}}(U) = \left\{ (f_p \in \mathcal{F}_p)_{p \in U} \mid \begin{array}{l} \text{for all } p \in U \text{ there exists a neighborhood } p \in V \subseteq U \text{ and } s \in \mathcal{F}(V) \\ \text{such that } s_q = f_q \text{ for all } q \in V \end{array} \right\},$$

where by s_q we mean the germ of s in $\mathcal{F}_q$. For $U \subseteq V$ there is an obvious map $\mathcal{F}^{\text{sh}}(V) \rightarrow \mathcal{F}^{\text{sh}}(U)$ mapping $(f_p)_{p \in V} \mapsto (f_p)_{p \in U}$; it is easily seen that $\mathcal{F}^{\text{sh}}$ is a presheaf.

Note that the construction of $\mathcal{F}^{\text{sh}}(U)$ is just the set of all families of compatible germs over U , which we discussed in the previous few results.

Now let $U \subseteq X$ be open and $\{U_i\}_i$ an open cover of U . Then if we have $(f_p^i)_{p \in U_i} \in \mathcal{F}^{\text{sh}}(U_i)$ for all i which are compatible, this means $f_p^i = f_p^j$ for all $p \in U_i \cap U_j$. Then we must define $(f_p)_{p \in U}$ by $f_p = f_p^i$ for any $i \in U_i$ for which p is a member of. This is well-defined precisely by compatibility, and it is a member of $\mathcal{F}^{\text{sh}}(U)$ because for $p \in U$ there is a $p \in U_i \subseteq U$ and then there is a $p \in V \subseteq U_i$ and $s \in \mathcal{F}(V)$ such that $s_q = f_q^i = f_q$ for all $q \in V$ as desired. Thus $\mathcal{F}^{\text{sh}}$ is a sheaf.

Now we must demonstrate that $\mathcal{F}^{\text{sh}}$ satisfies the universal property of a left adjoint. There is an obvious map $\text{sh}: \mathcal{F} \rightarrow \mathcal{F}^{\text{sh}}$ which maps $s \in \mathcal{F}(U)$ to $(s_p \in \mathcal{F}_p)_{p \in U} \in \mathcal{F}^{\text{sh}}(U)$ (this is a section because s satisfies the condition for it to be in $\mathcal{F}^{\text{sh}}(U)$). Let $g: \mathcal{F} \rightarrow \mathcal{G}$ be a morphism, and for $U \subseteq X$ consider the following diagram

$$\begin{array}{ccccc} \mathcal{F}(U) & \xrightarrow{\text{sh}} & \mathcal{F}^{\text{sh}}(U) & \xrightarrow{\iota_{\mathcal{F}}} & \prod_{p \in U} \mathcal{F}_p \\ & \searrow g & & & \downarrow \prod_p g_p \\ & & \mathcal{G}(U) & \xrightarrow{\iota_{\mathcal{G}}} & \prod_{p \in U} \mathcal{G}_p \end{array}$$

Note that the image of $\prod_p g_p \circ \iota_{\mathcal{F}}$ consists of compatible germs in $\prod_{p \in U} \mathcal{G}_p$ and so we can fill this diagram in with a unique arrow $f: \mathcal{F}^{\text{sh}}(U) \rightarrow \mathcal{G}(U)$. It is easy to see by uniqueness that these f assemble into a morphism $\mathcal{F}^{\text{sh}} \rightarrow \mathcal{G}$. ■

Note:

In the language of category theory, this theorem says that $\mathbf{Set}_X \subseteq \mathbf{Set}_X^{\text{pre}}$ is a **reflective subcategory**. Note that this proof works for other concrete categories like $\mathbf{Ab}$, $\mathbf{Grp}$, $\mathbf{Ring}$, etc. It can be shown that for sufficiently nice $\mathcal{C}$ (e.g. complete and cocomplete), sheafification exists for $\mathcal{C}_X \subseteq \mathcal{C}_X^{\text{pre}}$.

This justifies the fact that inclusion $\mathcal{C}_X \subseteq \mathcal{C}_X^{\text{pre}}$ preserves limits: it is a right adjoint. This explains why the presheaf kernel of a sheaf morphism is a sheaf kernel for example.

Proposition 1.3.7

The canonical morphism $\mathcal{F} \rightarrow \mathcal{F}^{\text{sh}}$ induces an isomorphism on stalks.

Proof

This is easily seen by an explicit description of $\mathcal{F}^{\text{sh}}$'s stalks. ■

Remark 1.3.8 (The space of sections of a presheaf)

Suppose $\mathcal{F}$ is a presheaf of sets on X . We will construct a topological space F along with a continuous map $\pi: F \rightarrow X$ as follows. As a set define F to be the disjoint unions of $\mathcal{F}$'s stalks:

$$F = \coprod_{p \in X} \mathcal{F}_p.$$

Now for $s \in \mathcal{F}(U)$ let $F_s = \{(p, s_p) \mid p \in U\} \subseteq F$; then the collection of all F_s where $s \in \mathcal{F}(U)$ for arbitrary $U \subseteq X$ forms a base for a topology on F . Indeed, for $s \in \mathcal{F}(U)$ and $t \in \mathcal{F}(V)$ we see that $F_s \cap F_t$ is the union of all F_r where $r = s|_W = t|_W$ for all $W \subseteq U \cap V$ where $s|_W = t|_W$.

The map $\pi: F \rightarrow X$ is the obvious projection $(p, s_p) \mapsto p$. For $U \subseteq X$ open we have that $\pi^{-1}U$ is the union of all F_s where $s \in \mathcal{F}(U)$ so that π is continuous. Furthermore π is an open map: for $s \in \mathcal{F}(U)$ it maps F_s to U . Moreover $\pi|_{F_s}$ is a bijection, so that π is a local homeomorphism.

The space F is called the **space of sections**, or **étalé space** of $\mathcal{F}$.

We can give an alternative construction for the sheafification of $\mathcal{F}$ as the sheaf of sections of $\pi: F \rightarrow X$ (that is for $U \subseteq X$ we define $\mathcal{F}^{\text{sh}}(U)$ to be the set of continuous maps $\sigma: U \rightarrow F$ where $\pi \circ \sigma = \text{id}_U$). Let us denote by $\mathcal{F}^+$ this new construction.

For $U \subseteq X$ open we define $\varphi: \mathcal{F}^{\text{sh}}(U) \rightarrow \mathcal{F}^+(U)$ by mapping $(f_p)_{p \in U}$ to $\sigma: U \rightarrow F$, $p \mapsto f_p$. Its inverse is given by mapping σ to $(\sigma(p))_p$.

Example 1.3.9

Let S be a set and consider the constant presheaf $\underline{S}_{\text{pre}}$. Then its sheafification is the constant *sheaf*: $(\underline{S}_{\text{pre}})^{\text{sh}} = \underline{S}$.

We can show this with the universal property: $\text{sh}: \underline{S}_{\text{pre}} \rightarrow \underline{S}$ is given by inclusion (constant maps are locally constant). Then for a morphism $g: \underline{S}_{\text{pre}} \rightarrow \mathcal{F}$ where $\mathcal{F}$ is a sheaf, we define $\bar{g}: \underline{S} \rightarrow \mathcal{F}$ by mapping a locally constant $f: U \rightarrow S$ to the gluing of $(\bar{g}(f|_{U_i}))_i$ (where $U_i \subseteq U$ are its connected components so that $f|_{U_i}$ is a constant).

Let $\varphi: \mathcal{F} \rightarrow \mathcal{G}$ be a morphism of sheaves, then recall the following equivalences (see remark 1.2.5 and proposition 1.3.5):

- (1) φ is monic as a morphism of sheaves,
- (2) φ is monic as a morphism of presheaves,

- (3) $\varphi(U)$ is monic for all $U \subseteq X$,
- (4) φ_p is monic for all $p \in X$.

A similar equivalence does not hold true for epimorphisms. This boils down to the fact that inclusion $\mathcal{C}_X \subseteq \mathcal{C}_X^{\text{pre}}$ does not necessarily preserve colimits, and thus not epimorphisms. So while a presheaf morphism is epic iff its components are, the same does not hold for sheaf morphisms. But it is true that epimorphisms are determined by stalks.

Proposition 1.3.10

Let $\varphi: \mathcal{F} \rightarrow \mathcal{G}$ be a morphism of sheaves. Then it is an epimorphism (in the category of sheaves) iff $\varphi_p: \mathcal{F}_p \rightarrow \mathcal{G}_p$ is an epimorphism for all $p \in X$.

Proof

If φ_p is epic for all p , then suppose we have two sheaf morphisms $\psi_1, \psi_2: \mathcal{G} \rightarrow \mathcal{H}$ such that $\psi_1 \circ \varphi = \psi_2 \circ \varphi$. Then taking stalks we see that for all p , $(\psi_1)_p \circ \varphi_p = (\psi_2)_p \circ \varphi_p$. Since φ_p is epic this means $(\psi_1)_p = (\psi_2)_p$ for all p and by proposition 1.3.4 this means $\psi_1 = \psi_2$ so that φ is epic.

Conversely let $p \in X$ and suppose there are maps $f, g: \mathcal{G}_p \rightarrow X$ such that $f \circ \varphi_p = g \circ \varphi_p$. We can define maps $\bar{f}, \bar{g}: \mathcal{G} \rightarrow p_*X$ by $\bar{f}_U(s) = f(s_p)$ for $p \in U$ and $\bar{f}_U = *$ otherwise. It is easy to see that $\bar{f} \circ \varphi = \bar{g} \circ \varphi$ and so $\bar{f} = \bar{g}$ which can easily be shown to mean $f = g$. ■

Note:

In remark 1.6.2 will show that there is an adjunction between taking stalks and the skyscraper sheaf: $\bullet_p \dashv p_* \bullet$. This was used in the above proof, and it is easy to see then how to construct this.

Example 1.3.11

We will show that an epimorphism of sheaves need not be epic on each of its components.

Let $\mathcal{O}_{\mathbb{C}}$ be the sheaf of holomorphic functions and $\mathcal{O}_{\mathbb{C}}^*$ be the sheaf of invertible (nowhere zero) holomorphic functions. These are sheaves of Abelian groups, and we then have maps of sheaves

$$0 \rightarrow \mathbb{Z} \xrightarrow{\times 2\pi i} \mathcal{O}_{\mathbb{C}} \xrightarrow{\exp} \mathcal{O}_{\mathbb{C}}^* \rightarrow 1.$$

We will soon show that this is an exact sequence of sheaves of Abelian groups.

In particular $\exp: \mathcal{O}_{\mathbb{C}} \rightarrow \mathcal{O}_{\mathbb{C}}^*$ is an epimorphism as we can show directly. For any $z \in \mathbb{C}$ the morphism of stalks $\exp_z$ is surjective because every invertible holomorphic function locally has a logarithm. But $\exp$ is not an epimorphism on each of its components, for example $\text{id} \in \mathcal{O}_{\mathbb{C}}^*(\mathbb{C} - 0)$ has no global logarithm.

1.4 B-Sheaves

Oftentimes it is more natural to define a sheaf on X by defining its sections on a base of X , instead of on every open subset of X . Recall that a base of a topology is a collection of open subsets $\mathcal{B} \subseteq \text{open}(X)$ such that every open subset of X can be written as a union of sets from $\mathcal{B}$.

In the absence of an ambient topology, we may mean a base to mean a collection $\mathcal{B}$ of subsets of X such that the intersection of any two sets from $\mathcal{B}$ can be written as a union of elements from $\mathcal{B}$. If we are given an arbitrary collection S of subsets of X , they generate a minimal topology on X ; S is called a **subbase** of this topology. The topology generated by $\mathcal{B}$ consists of all finite intersections of elements from $\mathcal{B}$, and $\mathcal{B}$ is a base of this topology (in the first sense).

As $\mathcal{B}$ is a collection of subsets of X , we can view it as a poset under inclusion, and thus a subcategory of $\text{open}(X)$.

Definition 1.4.1 (B-Sheaves)

Let X be a topological space with a given base $\mathcal{B}$. Define a **$\mathcal{B}$ -presheaf** to be a functor $\mathcal{F}: \mathcal{B}^{\text{op}} \rightarrow \text{Set}$. This presheaf is a **$\mathcal{B}$ -sheaf** if it satisfies the following two axioms:

- (1) the **identity axiom**: if $B = \bigcup_i B_i$ is a basic covering and $f, g \in \mathcal{F}(B)$ such that $f|_{B_i} = g|_{B_i}$ for all i , then $f = g$.
- (2) the **gluing axiom**: if $B = \bigcup_i B_i$ is a basic covering and $f_i \in \mathcal{F}(B_i)$ are given for all i such that $f_i|_{B_k} = f_j|_{B_k}$ for all $B_k \subseteq B_i \cap B_j$, then there is an $f \in \mathcal{F}(B)$ such that $f|_{B_i} = f_i$ for all i .

We can rewrite the following axioms categorically as follows. For $B = \bigcup_i B_i$, the following diagram must be an equalizer map

$$\mathcal{F}(B) \xrightarrow{\varphi} \prod_i \mathcal{F}(B_i) \xrightarrow{\alpha} \prod_{i,j,k} \mathcal{F}(B_{i,j,k})$$

where $\{B_{i,j,k}\}_k$ ranges over all basic sets contained in $B_i \cap B_j$, and

$$\theta = (\text{res}_{B, B_i})_i, \quad \alpha = (\text{res}_{B_i, B_k} \circ \text{proj}_i)_{i,j,k}, \quad \beta = (\text{res}_{B_j, B_k} \circ \text{proj}_j)_{i,j,k}.$$

Thus we can define $\mathcal{B}$ -sheaves over any category $\mathcal{C}$ with enough limits.

Definition 1.4.2

Denote the category of $\mathcal{C}$ -valued $\mathcal{B}$ -presheaves by $\mathcal{C}_B^{\text{pre}}$, and the category of $\mathcal{C}$ -valued $\mathcal{B}$ -sheaves by $\mathcal{C}_B$.

Theorem 1.4.3

Let $\mathcal{B}$ be a base for a topology on X , and $\mathcal{F}$ a sheaf of sets on this base. Then there is a unique sheaf of sets $\mathcal{F}^{\text{ext}}$ on X extending $\mathcal{F}$.

This extends to an equivalence of categories $\text{ext}: \text{Set}_B \xrightarrow{\sim} \text{Set}_X$; res , where $\text{res}: \text{Set}_X \rightarrow \text{Set}_B$ is the obvious restriction functor.

Proof

For a $\mathcal{B}$ -sheaf we can define stalks much the same way as for a usual sheaf. For $p \in X$ define

$$\mathcal{F}_p = \text{colim}_{p \in B \in \mathcal{B}} \mathcal{F}(B).$$

Then define for $U \subseteq X$,

$$\mathcal{F}^{\text{ext}}(U) = \left\{ (f_p \in \mathcal{F}_p)_{p \in U} \mid \begin{array}{l} \text{for all } p \in U \text{ there is a } B \in \mathcal{B} \text{ with } p \in B \subseteq U, \\ \text{such that } s_q = f_q \text{ for all } q \in B \end{array} \right\}.$$

There are obvious restriction maps and it is easy to see that this is a sheaf. Furthermore it is easy to see that $\mathcal{F}^{\text{ext}}(B)$ is the image of $\mathcal{F}(B)$ under $\iota_{\mathcal{F}}$ which is an injection as for usual sheaves. $\blacksquare$

Remark 1.4.4

Suppose we have an open cover $X = \bigcup_i U_i$ and for each i a sheaf of sets $\mathcal{F}_i$ on U_i . Furthermore suppose we have isomorphisms

$$\varphi_{ij}: \mathcal{F}_i|_{U_i \cap U_j} \xrightarrow{\sim} \mathcal{F}_j|_{U_i \cap U_j}$$

such that φ_{ii} is the identity and

$$\varphi_{jk} \circ \varphi_{ij} = \varphi_{ik}$$

for all i, j, k on $U_i \cap U_j \cap U_k$.

There then exists a unique sheaf $\mathcal{F}$ on X such that $\mathcal{F}|_{U_i}$ is isomorphic to $\mathcal{F}_i$.

This is because we can define a base $\mathcal{B}$ consisting of those open subsets contained in some U_i , i.e. $\mathcal{B} = \bigcup_i \text{open}(U_i)$. Then there is a unique way to extend the $\mathcal{F}_i$ s to a $\mathcal{B}$ -sheaf. For $U \subseteq U_i$ we define $\mathcal{F}(U) = \mathcal{F}_i(U)$ (this is well-defined because if $U \subseteq U_i \cap U_j$ then $\mathcal{F}_i(U) \cong \mathcal{F}_j(U)$), and the restriction maps are given by the restriction maps of $\mathcal{F}_i$ and the φ_{ij} . That is, for $U \subseteq U_i$ and $V \subseteq U_j$ where $U \subseteq V$ we define $\text{res}: \mathcal{F}_j(V) \rightarrow \mathcal{F}_i(U)$ by

$$\text{res} = \mathcal{F}_j(V) \xrightarrow{\text{res}_j} \mathcal{F}_j(U) \xrightarrow{\varphi_{ji}} \mathcal{F}_i(U).$$

The conditions on φ_{ij} ensure that this is a $\mathcal{B}$ -sheaf, and we can extend it to a sheaf on X via the above theorem.

1.5 Abelian categories of sheaves

Recall that for a sheaf of rings $\mathcal{O}_X$, an $\mathcal{O}_X$ -**module** is a sheaf of Abelian groups $\mathcal{F}$ such that each $\mathcal{F}(U)$ is an $\mathcal{O}_X(U)$ -module in a compatible way (definition 1.1.12). A sheaf of Abelian groups is a $\mathbb{Z}$ -module, so $\mathcal{O}_X$ -modules generalize the concept of sheaves of Abelian groups much like how modules generalize the concept of Abelian groups.

The goal of this section is to demonstrate that the category of $\mathcal{O}_X$ -modules (and in particular sheaves of Abelian groups) forms an Abelian category.

To do this we need to show that the category of $\mathcal{O}_X$ -modules has kernels and cokernels, and that monics are kernels and epics are cokernels. Firstly let us consider $\text{Mod}_{\mathcal{O}_X}^{\text{pre}}$, the category of presheaf $\mathcal{O}_X$ -modules. The definition here is the same as definition 1.1.12 except we require only that $\mathcal{F}$ be a presheaf.

Now $\text{Mod}_{\mathcal{O}_X}^{\text{pre}}$ is complete and cocomplete. If $\Phi: I \rightarrow \text{Mod}_{\mathcal{O}_X}^{\text{pre}}$ is a diagram, then there is an obvious forgetful functor $\text{Mod}_{\mathcal{O}_X}^{\text{pre}} \rightarrow \text{Ab}_X^{\text{pre}}$. Composing this with Φ we can take the co/limit, and then there is an obvious $\mathcal{O}_X$ -module structure on the resulting object (this is the same situation as with $\text{Mod}_R \rightarrow \text{Ab}$).

Then sheafification lifts to $\text{Mod}_{\mathcal{O}_X}^{\text{pre}} \rightarrow \text{Mod}_{\mathcal{O}_X}$. This is clear by considering $\mathcal{F}^{\text{sh}}(U)$ explicitly. Thus $\text{Mod}_{\mathcal{O}_X}$ is a reflective subcategory of $\text{Mod}_{\mathcal{O}_X}^{\text{pre}}$, and since the latter is co/complete so too is the former.

In particular the kernel of a $\text{Mod}_{\mathcal{O}_X}$ -morphism is the kernel of it viewed as a morphism in Ab_X^{pre} , and its cokernel is the sheafification of this morphism.

Recall that the stalk of a morphism in $\text{Mod}_{\mathcal{O}_X}$ is an $\mathcal{O}_{X,p}$ -module. Thus we view $\bullet_p$ as a functor

$$\bullet_p: \text{Mod}_{\mathcal{O}_X} \rightarrow \text{Mod}_{\mathcal{O}_{X,p}}.$$

When $\mathcal{O}_X = \mathbb{Z}$ then $\text{Mod}_{\mathcal{O}_X}$ is the same as Ab_X , and $\mathcal{O}_{X,p}$ is the same as $\mathbb{Z}$ so that $\text{Mod}_{\mathcal{O}_{X,p}} \cong \text{Ab}$. Thus all of the following results hold for sheaves of Abelian groups.

Lemma 1.5.1

Taking stalks preserves kernels and cokernels: for $p \in X$ and a morphism $\varphi: \mathcal{F} \rightarrow \mathcal{G}$ of $\mathcal{O}_X$ -modules,

$$(\ker \varphi)_p \cong \ker(\varphi_p), \quad (\text{coker } \varphi)_p \cong \text{coker}(\varphi_p).$$

Proof

This is clear for cokernels, as the stalk at p amounts to taking a certain colimit, and colimits (such as cokernels) commute with colimits.

For kernels we can show this directly. We view both $(\ker \varphi)_p$ and $\ker(\varphi_p)$ canonically as subobjects of $\mathcal{F}_p$, in which case it is clear that there is an equality. ■

Lemma 1.5.2

In $\mathbf{Mod}_{\mathcal{O}_X}$, a morphism $\varphi: \mathcal{F} \rightarrow \mathcal{G}$ is monic iff its stalks are monic, epic iff its stalks are epic, and an isomorphism iff its stalks are isomorphisms.

Note:

It doesn't matter if we view φ_p as existing in $\mathbf{Mod}_{\mathcal{O}_{X,p}}$ or $\mathbf{Set}$; a module morphism (in the ordinary sense) is a monomorphism/epimorphism/isomorphism iff it is injective/surjective/bijective.

Note that in lieu of this, a $\mathbf{Mod}_{\mathcal{O}_X}$ -morphism with any of the above properties has those properties in $\mathbf{Ab}_X$ and $\mathbf{Set}_X$.

Proof

First we note that stalks determine a morphism: if $\varphi_p = \psi_p$ for all $p \in X$ then $\varphi = \psi$. This is because the forgetful functor $U: \mathbf{Mod}_{\mathcal{O}_X} \rightarrow \mathbf{Set}_X$ is faithful, and commutes with taking stalks. So $U\varphi_p = U\psi_p$, which implies $U\varphi = U\psi$ by proposition 1.3.4, and thus $\varphi = \psi$.

Then the proofs of proposition 1.3.5 and proposition 1.3.10 are valid here. $\blacksquare$

This allows us to prove the desired result.

Theorem 1.5.3

$\mathbf{Mod}_{\mathcal{O}_X}$ is an Abelian category.

Proof

As already shown, $\mathbf{Mod}_{\mathcal{O}_X}$ is co/complete. Furthermore it is clearly a pre-additive category. We need only show that for a monic φ and an epic ψ ,

$$\varphi = \ker \operatorname{coker} \varphi, \quad \psi = \operatorname{coker} \ker \psi.$$

But this is true only if for all $p \in X$,

$$\varphi_p = \ker \operatorname{coker} \varphi_p, \quad \psi_p = \operatorname{coker} \ker \psi_p \tag{1}$$

(parentheses on the left-hand sides of the equations here are unnecessary, as per lemma 1.5.1). By lemma 1.5.2 φ_p, ψ_p are monic and epic respectively, so (1) holds as $\mathbf{Mod}_{\mathcal{O}_{X,p}}$ is Abelian. $\blacksquare$

Note:

It is actually the case that if $\mathcal{A}$ is an arbitrary (complete) Abelian category, then $\mathcal{A}_X$ is Abelian as well.

Lemma 1.5.4

Let φ be a $\mathbf{Mod}_{\mathcal{O}_X}$ -morphism, then $\operatorname{im} \varphi$ is the sheafification of the image presheaf of φ . In particular $(\operatorname{im} \varphi)_p = \operatorname{im}(\varphi_p)$.

Proof

We know that $\operatorname{im}_{\operatorname{pre}} \varphi = \ker_{\operatorname{pre}} \operatorname{coker}_{\operatorname{pre}} \varphi$, while $\operatorname{im} \varphi = \ker \operatorname{coker} \varphi$. Since $\operatorname{coker} = \operatorname{coker}_{\operatorname{pre}}^{\operatorname{sh}}$ and $\ker = \ker_{\operatorname{pre}}^{\operatorname{sh}} = \ker_{\operatorname{pre}} \operatorname{so}$

$$\operatorname{im} \varphi = \ker_{\operatorname{pre}}^{\operatorname{sh}} \operatorname{coker}_{\operatorname{pre}}^{\operatorname{sh}} \varphi = (\ker_{\operatorname{pre}} \operatorname{coker}_{\operatorname{pre}} \varphi)^{\operatorname{sh}} = (\operatorname{im}_{\operatorname{pre}} \varphi)^{\operatorname{sh}}$$

as desired.

The second point follows from stalks commuting with kernels and cokernels. ■

Corollary 1.5.5

The following sequence of $\operatorname{Mod}_{\mathcal{O}_X}$ -morphisms

$$\mathcal{F} \xrightarrow{\alpha} \mathcal{G} \xrightarrow{\beta} \mathcal{H}$$

is exact iff

$$\mathcal{F}_p \xrightarrow{\alpha_p} \mathcal{G}_p \xrightarrow{\beta_p} \mathcal{H}_p$$

is exact for all $p \in X$.

Proof

The former is exact iff $\operatorname{im} \alpha = \ker \beta$ iff $(\operatorname{im} \alpha)_p = (\ker \beta)_p$ for all $p \in X$, which happens iff $\operatorname{im} \alpha_p = \ker \beta_p$, i.e. if the latter sequence is exact. ■

An immediate consequence of this corollary is the following.

Corollary 1.5.6

Taking stalks is an exact functor $\bullet_p: \operatorname{Mod}_{\mathcal{O}_X} \rightarrow \operatorname{Mod}_{\mathcal{O}_{X,p}}$.

Example 1.5.7

For $U \subseteq X$, there is a functor $\Gamma_U: \mathcal{C}_X \rightarrow \mathcal{C}$ mapping a sheaf $\mathcal{F} \mapsto \mathcal{F}(U)$. It is easy to see that $\Gamma_U: \operatorname{Mod}_{\mathcal{O}_X} \rightarrow \operatorname{Mod}_{\mathcal{O}_X(U)}$ is a left-exact functor. That is, it preserves kernels. This is clear because the kernel of a morphism of sheaves $\mathcal{F} \rightarrow \mathcal{G}$ is just the presheaf kernel, which is taken coordinate-wise.

When $U = X$ we denote this by just $\Gamma = \Gamma_X$, and it is called the **global sections functor**. Note that there is an obvious restriction functor $\operatorname{res}: \mathcal{C}_X \rightarrow \mathcal{C}_U$ mapping a sheaf $\mathcal{F}$ to its restriction $\mathcal{F}|_U$. Then Γ_U is equal to the composition

$$\Gamma_U = \mathcal{C}_X \xrightarrow{\operatorname{res}} \mathcal{C}_U \xrightarrow{\Gamma} \mathcal{C}.$$

And it is easy to see that res is also exact.

It is not true that Γ_U is exact though. Consider the exact sequence

$$0 \rightarrow \underline{\mathbb{Z}} \xrightarrow{\times 2\pi i} \mathcal{O}_{\mathbb{C}} \xrightarrow{\exp} \mathcal{O}_{\mathbb{C}}^* \rightarrow 1$$

of example 1.3.11. As before, $\Gamma_U(\exp)$ is not surjective as not every nowhere zero holomorphic function has a global inverse. Take for example $U = \mathbb{C} - 0$ and the function id .

Note:

Note that the maps $\text{res}: \mathcal{C}_V \rightarrow \mathcal{C}_U$ for $U \subseteq V$ forms a presheaf $\text{open}(X)^{\text{op}} \rightarrow \mathbf{Cat}$. In fact it is a $\mathbf{Cat}$ -valued sheaf.

Note:

We will show soon that $\Gamma: \mathcal{C}_X \rightarrow \mathcal{C}$ is right-adjoint to the constant sheaf functor. And as we will see $\text{res}: \mathcal{C}_X \rightarrow \mathcal{C}_U$ is the inverse image sheaf of the inclusion $U \hookrightarrow X$, which in general is left-adjoint to the pushforward, but in the case of an open embedding it is a right-adjoint as well. Thus $\Gamma_U = \Gamma \circ \text{res}$ is a right-adjoint and thus preserves limits; in particular it is left-exact.

Example 1.5.8

As alluded to in the previous note, the pushforward functor $\pi_*: \mathcal{C}_X \rightarrow \mathcal{C}_Y$ associated to a continuous map $\pi: X \rightarrow Y$ is a right-adjoint, and thus left-exact. We can show left-exactness directly: for a morphism $\varphi: \mathcal{F} \rightarrow \mathcal{G}$ in $\mathcal{C}_X$, the components of $\pi_*\varphi$ are $(\pi_*\varphi)_V = \varphi_{f^{-1}V}$ for $V \subseteq Y$. Because kernels are taken coordinate-wise:

$$(\ker \pi_*\varphi)_V = \ker(\pi_*\varphi)_V = \ker \varphi_{f^{-1}V} = (\ker \varphi)_{f^{-1}V} = (\pi_* \ker \varphi)_V$$

so that $\ker \pi_*\varphi = \pi_* \ker \varphi$ as desired.

Definition 1.5.9

Let $\mathcal{F}, \mathcal{G}$ be $\mathcal{O}_X$ -modules. Then we form their **tensor product presheaf** to be the presheaf $\mathcal{F} \otimes_{\mathcal{O}_X}^{\text{pre}} \mathcal{G}$ given by

$$(\mathcal{F} \otimes_{\mathcal{O}_X}^{\text{pre}} \mathcal{G})(U) = \mathcal{F}(U) \otimes_{\mathcal{O}_X(U)} \mathcal{G}(U), \quad \text{res}_{V,U} = \text{res}_{V,U}^{\mathcal{F}} \otimes \text{res}_{V,U}^{\mathcal{G}}.$$

Their **tensor product** is the sheafification of the tensor product presheaf:

$$\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G} = (\mathcal{F} \otimes_{\mathcal{O}_X}^{\text{pre}} \mathcal{G})^{\text{sh}}.$$

Proposition 1.5.10

For a presheaf $\mathcal{F}$, $\bullet \otimes^{\text{pre}} \mathcal{F}$ is left adjoint to $\mathbf{Hom}(\mathcal{F}, \bullet)$ as functors $\mathbf{Mod}_{\mathcal{O}_X}^{\text{pre}} \rightarrow \mathbf{Mod}_{\mathcal{O}_X}^{\text{pre}}$.

If $\mathcal{F}$ is a sheaf then $\bullet \otimes \mathcal{F}$ is left adjoint to $\mathbf{Hom}(\mathcal{F}, \bullet)$ as functors $\mathbf{Mod}_{\mathcal{O}_X} \rightarrow \mathbf{Mod}_{\mathcal{O}_X}$.

Furthermore, $\mathbf{Hom}(\bullet, \mathcal{F})$ is a self-adjoint functor. That is, there is an adjunction $\mathbf{Hom}(\bullet, \mathcal{F})^{\text{op}}: \mathbf{Mod}_{\mathcal{O}_X}^{\text{op}} \rightleftarrows \mathbf{Mod}_{\mathcal{O}_X}: \mathbf{Hom}(\bullet, \mathcal{F})$.

Proof

If $\varphi: \mathcal{G} \otimes^{\text{pre}} \mathcal{F} \rightarrow \mathcal{H}$ is a morphism, then for all $U \subseteq X$, $\varphi_U: \mathcal{G}(U) \otimes \mathcal{F}(U) \rightarrow \mathcal{H}(U)$ corresponds to a map $\psi_U: \mathcal{G}(U) \rightarrow \mathbf{Hom}_{\mathcal{O}_X(U)}(\mathcal{F}(U), \mathcal{H}(U))$. This then defines a map $\widehat{\varphi}_U: \mathcal{G}(U) \rightarrow \mathbf{Hom}(\mathcal{F}, \mathcal{H})(U)$ which maps $x \in \mathcal{G}(U)$ to $\widehat{\varphi}_U(x): \mathcal{F}|_U \rightarrow \mathcal{H}|_U$ whose components on $V \subseteq U$ are ψ_V .

Conversely for a map $\psi: \mathcal{G} \rightarrow \mathbf{Hom}(\mathcal{F}, \mathcal{H})$, we get maps $\mathcal{G}(U) \rightarrow \mathbf{Hom}(\mathcal{F}|_U, \mathcal{H}|_U) \xrightarrow{\Gamma} \mathbf{Hom}(\mathcal{F}(U), \mathcal{H}(U))$ which then gives us a map $\widehat{\psi}_U: \mathcal{G}(U) \otimes \mathcal{F}(U) \rightarrow \mathcal{H}(U)$. This assembles into the desired map $\widehat{\psi}: \mathcal{G} \otimes^{\text{pre}} \mathcal{F} \rightarrow \mathcal{H}$.

Thus we have adjunctions

$$\mathbf{Mod}_{\mathcal{O}_X}^{\text{pre}} \xrightleftharpoons[\mathbf{Hom}(\mathcal{F}, \bullet)]{\otimes^{\text{pre}} \mathcal{F}} \mathbf{Mod}_{\mathcal{O}_X}^{\text{pre}} \xrightleftharpoons[\text{inc}]{\text{sh}} \mathbf{Mod}_{\mathcal{O}_X}$$

so that $\bullet \otimes \mathcal{F}$ is the left adjoint of $\mathbf{Hom}(\mathcal{F}, \bullet)$ as desired.

For the final point we note that the adjunctions above are natural in all variables, so that

$$\mathrm{hom}(\mathcal{H}, \mathbf{Hom}(\mathcal{G}, \mathcal{F})) \cong \mathrm{hom}(\mathcal{H} \otimes \mathcal{G}, \mathcal{F}) \cong \mathrm{hom}(\mathcal{H} \otimes \mathcal{G}, \mathcal{F}) \cong \mathrm{hom}(\mathcal{H}, \mathbf{Hom}(\mathcal{G}, \mathcal{F}))$$

as desired. ■

Corollary 1.5.11

For a sheaf $\mathcal{F}$, $\mathbf{Hom}(\mathcal{F}, \bullet)$ is left-exact and $\mathbf{Hom}(\bullet, \mathcal{F})$ is contravariant left-exact.

Proposition 1.5.12

If $\mathcal{F}, \mathcal{G}$ are $\mathcal{O}_X$ -modules then for $p \in X$,

$$(\mathcal{F} \otimes \mathcal{G})_p = \mathcal{F}_p \otimes \mathcal{G}_p.$$

Proof

Filtered colimits in module categories commute with tensor products so

$$(\mathcal{F} \otimes \mathcal{G})_p = \mathrm{colim}_{p \in U} (\mathcal{F}(U) \otimes \mathcal{G}(U)) = (\mathrm{colim}_U \mathcal{F}(U)) \otimes (\mathrm{colim}_U \mathcal{G}(U)) = \mathcal{F}_p \otimes \mathcal{G}_p. \quad \blacksquare$$

1.6 Adjunctions of sheaves

In the following section, $\mathcal{C}$ is a complete and cocomplete category (or at least has all filtered colimits) for which sheafification exists.

Recall that for a continuous map $\pi: X \rightarrow Y$ there is a functor $\pi_*: \mathcal{C}_X \rightarrow \mathcal{C}_Y$ mapping a sheaf $\mathcal{F}$ to the sheaf $(\pi_*\mathcal{F})(V) = \mathcal{F}(\pi^{-1}V)$ for $V \subseteq Y$. It turns out that this functor has a left adjoint.

Theorem 1.6.1

The pushforward functor $\pi_*: \mathcal{C}_X \rightarrow \mathcal{C}_Y$ has a left adjoint, called the **inverse image functor**, $\pi^{-1}: \mathcal{C}_Y \rightarrow \mathcal{C}_X$.

Proof

First we show that $\pi_*: \mathcal{C}_X^{\mathrm{pre}} \rightarrow \mathcal{C}_Y^{\mathrm{pre}}$ has a left adjoint. For $U \subseteq X$ and $\mathcal{G} \in \mathcal{C}_Y^{\mathrm{pre}}$ define

$$\pi_{\mathrm{pre}}^{-1}\mathcal{G}(U) = \mathrm{colim}_{V \supseteq \pi(U)} \mathcal{G}(V).$$

This is clearly functorial in U and thus defines a presheaf in $\mathcal{C}_X^{\mathrm{pre}}$. Furthermore it is also clearly functorial in $\mathcal{G}$, defining a functor $\mathcal{C}_Y^{\mathrm{pre}} \rightarrow \mathcal{C}_X^{\mathrm{pre}}$.

To show the adjunction, consider we have a morphism $\varphi: \pi^{-1}\mathcal{G} \rightarrow \mathcal{F}$, which has components

$$\varphi_U: \mathrm{colim}_{V \supseteq \pi(U)} \mathcal{G}(V) \rightarrow \mathcal{F}(U).$$

Then define $\widehat{\varphi}: \mathcal{G} \rightarrow \pi_*\mathcal{F}$ whose component on $V \subseteq Y$ is

$$\widehat{\varphi}_V = \mathcal{G}(V) \rightarrow \mathrm{colim}_{W \supseteq \pi(\pi^{-1}V)} \mathcal{G}(W) \xrightarrow{\varphi_{\pi^{-1}V}} \mathcal{F}(\pi^{-1}V),$$

where the first morphism comes from the colimit cone.

Conversely given $\psi: \mathcal{G} \rightarrow \pi_* \mathcal{F}$ then for $U \subseteq X$ and $V \supseteq \pi U$ there is a morphism

$$\mathcal{G}(V) \xrightarrow{\psi_V} \mathcal{F}(\pi^{-1}V) \xrightarrow{\text{res}} \mathcal{F}(U).$$

These assemble into a cone under $(V \supseteq \pi U) \mapsto \mathcal{G}(V)$, and thus defines a map $\tilde{\psi}_U: \text{colim}_{V \supseteq \pi U} \mathcal{G}(V) \rightarrow \mathcal{F}(U)$. These assemble into a morphism $\tilde{\psi}: \pi^{-1} \mathcal{G} \rightarrow \mathcal{F}$.

Now for sheaves, we have adjunctions

$$\mathcal{C}_Y^{\text{pre}} \xrightleftharpoons[\pi_*]{\pi_{\text{pre}}^{-1}} \mathcal{C}_X^{\text{pre}} \xrightleftharpoons[\text{inc}]{\text{sh}} \mathcal{C}_X$$

so we define $\pi^{-1} = \text{sh} \circ \pi_{\text{pre}}^{-1}$ and this gives us the desired left adjoint. ■

Remark 1.6.2

A point $p \in X$ may be equivalently viewed as a morphism from the one-point space $p: * \rightarrow X$ which picks out p . Sheaves on $*$ amount to objects of $\mathcal{C}$ (because the sheaf condition implies $\mathcal{F}(\emptyset)$ is the terminal object), so that the inverse image-pushforward amounts to an adjunction

$$\mathcal{C}_X \xrightleftharpoons[p_*]{p^{-1}} \mathcal{C}.$$

When viewing $A \in \mathcal{C}$ as a sheaf over $*$, we get that $p_* A$ is the sheaf on X where for $U \subseteq X$,

$$(p_* A)(U) = A(p^{-1}U) = \begin{cases} A & p \in U \\ * & \text{else} \end{cases},$$

as A is the sheaf $A(*) = A$ and $A(\emptyset) = *$. This is precisely the skyscraper sheaf (which justifies the notation $p_* A$).

On the other hand, $p^{-1} \mathcal{F} \in \mathcal{C}$ is the colimit

$$\text{colim}_{U \ni p} \mathcal{F}(U) = \mathcal{F}_p$$

so that p^{-1} is the stalk functor at p .

So in this specific case, we have an adjunction $\bullet_p \dashv p_* \bullet$.

Proposition 1.6.3

The construction $\pi \mapsto \pi_*$ is functorial, and $\pi \mapsto \pi^{-1}$ is contravariantly functorial. That is, these form functors

$$\begin{aligned} \text{Top} &\rightarrow \text{Cat}, & X &\mapsto \mathcal{C}_X, & \pi &\mapsto \pi_* \\ \text{Top}^{\text{op}} &\rightarrow \text{Cat}, & X &\mapsto \mathcal{C}_X, & \pi &\mapsto \pi^{-1}. \end{aligned}$$

Proof

Because π^{-1} is the left adjoint of π_* , it is sufficient to show that one construction is functorial to show that the second is. We will show that if we have a composition of continuous maps $X \xrightarrow{\pi} Y \xrightarrow{\tau} Z$, then $(\tau \circ \pi)_* = \tau_* \circ \pi_*$. Indeed,

$$(\tau_* \circ \pi_* \mathcal{F})(W) = \pi_* \mathcal{F}(\tau^{-1}W) = \mathcal{F}((\tau \circ \pi)^{-1}W) = (\tau \circ \pi)_* \mathcal{F}(W). \quad \blacksquare$$

Note:

It should be made more clear what it is meant for $\pi \mapsto \pi^{-1}$ to be functorial. Indeed, it is not strictly true that $(\tau \circ \pi)^{-1} = \pi^{-1} \circ \tau^{-1}$; this equality is not strict and is rather a natural isomorphism. So $\pi \mapsto \pi^{-1}$ is not an ordinary functor, rather a 2-functor.

Corollary 1.6.4

Let $\pi: X \rightarrow Y$ be continuous, and $\mathcal{G} \in \mathcal{C}_Y$. Then for $p \in X$ there is a natural isomorphism $(\pi^{-1}\mathcal{G})_p \cong \mathcal{G}_{\pi p}$.

Proof

Note that $(\pi^{-1}\mathcal{G})_p$ is the composition $p^{-1} \circ \pi^{-1}(\mathcal{G}) = (\pi \circ p)^{-1}\mathcal{G}$, with $p: * \rightarrow X$. This is equal to $\pi(p)^{-1}(\mathcal{G}) = \mathcal{G}_{\pi p}$ as desired. ■

Proposition 1.6.5

Suppose $U \subseteq X$ is an open subset and consider the inclusion $\iota: U \hookrightarrow X$. Then the functors ι^{-1} and $\bullet|_U$ (restriction) are isomorphic.

Proof

We will show that $\bullet|_U$ is a left adjoint to ι_* . A map $\varphi: \mathcal{F}|_U \rightarrow \mathcal{G}$ defines $\widehat{\varphi}: \mathcal{F} \rightarrow \iota_*\mathcal{G}$ by

$$\widehat{\varphi}_V = \mathcal{F}(V) \xrightarrow{\text{res}} \mathcal{F}(U \cap V) \xrightarrow{\varphi|_{U \cap V}} \mathcal{G}(U \cap V) = \iota_*\mathcal{G}(V).$$

Conversely $\psi: \mathcal{F} \rightarrow \iota_*\mathcal{G}$ defines $\widetilde{\psi}: \mathcal{F}|_U \rightarrow \mathcal{G}$ by $\widetilde{\psi}_V = \psi_V$ for $V \subseteq U$, as $\iota_*\mathcal{G}(V) = \mathcal{G}(V)$. ■

Proposition 1.6.6

Let $\pi: X \rightarrow Y$ be continuous, then $\pi^{-1}: \mathbf{Ab}_Y \rightarrow \mathbf{Ab}_X$ is exact.

Proof

Exactness can be checked on the level of stalks by corollary 1.5.5. Thus π^{-1} is exact iff $p^{-1} \circ \pi^{-1}$ is exact for all $p \in X$ (where p^{-1} denotes taking the stalk at p). But this is isomorphic to $\pi(p)^{-1} = \bullet_{\pi p}$ which is exact. ■

Consider the following situation: we have a commutative square of continuous maps

$$\begin{array}{ccc} W & \xrightarrow{\beta'} & X \\ \alpha' \downarrow & & \downarrow \alpha \\ Y & \xrightarrow{\beta} & Z \end{array}$$

For a sheaf $\mathcal{F}$ on X we can define the **push-pull map**

$$\beta^{-1} \circ \alpha_* \mathcal{F} \rightarrow \alpha'_*(\beta')^{-1} \mathcal{F}$$

as follows. Consider the unit of $(\beta')^{-1} \dashv \beta'_*$, $\eta^{\beta'}: \mathcal{F} \rightarrow (\beta'_*)(\beta')^{-1}\mathcal{F}$. Then apply α_* to get $\alpha_*\mathcal{F} \rightarrow \alpha_*(\beta'_*)(\beta')^{-1}\mathcal{F}$. By commutativity this is equal to $\alpha_*\mathcal{F} \rightarrow (\beta_*)(\alpha'_*)(\beta')^{-1}\mathcal{F}$, and we can apply the adjunction $\beta^{-1} \dashv \beta$ to get $\beta^{-1}\alpha_*\mathcal{F} \rightarrow \alpha'_*(\beta')^{-1}\mathcal{F}$ as desired.

Recall that given an adjunction $(\eta, \varepsilon): L \dashv R$, the isomorphism $\text{hom}(c, Rd) \xrightarrow{\sim} \text{hom}(Lc, d)$ is given $f \mapsto \varepsilon \circ Lf$. Thus the push-pull map is given by

$$\varepsilon_{\alpha'_*(\beta')^{-1}}^{\beta} \circ (\beta^{-1} \circ \alpha_*(\eta^{\beta'})).$$

But we could define this another way. Instead we start with the counit $\varepsilon^{\alpha}: \alpha^{-1}\alpha_*\mathcal{F} \rightarrow \mathcal{F}$, apply $(\beta')^{-1}$: $(\beta')^{-1}\alpha^{-1}\alpha_*\mathcal{F} \rightarrow (\beta')^{-1}\mathcal{F}$, which is equivalent to $(\alpha')^{-1}\beta^{-1}\alpha_*\mathcal{F} \rightarrow (\beta')^{-1}\mathcal{F}$, then we apply the adjunction $(\alpha')^{-1} \dashv \alpha'_*$ to get the desired $\beta^{-1}\alpha_*\mathcal{F} \rightarrow \alpha'_*(\beta')^{-1}\mathcal{F}$. In this case the map becomes

$$(\alpha'_* \circ (\beta')^{-1})(\varepsilon^{\alpha}) \circ \eta^{\alpha'}.$$

These two definitions are equivalent. Indeed, we can view both of these maps as the $\mathcal{F}$ -component of a certain natural transformation, given by the two paths in the following diagram (the whiskering in the diagram is left implicit):

$$\begin{array}{ccc} \beta^{-1}\alpha_*\beta'_*(\beta')^{-1} & \xrightarrow{\sim} & \beta^{-1}\beta_*\alpha'_*(\beta')^{-1} \\ \eta^{\beta'} \uparrow & & \downarrow \varepsilon^{\beta} \\ \beta^{-1}\alpha_* & & \alpha'_*(\beta')^{-1} \\ \eta^{\alpha'} \downarrow & & \uparrow \varepsilon^{\alpha} \\ \alpha'_*(\alpha')^{-1}\beta^{-1}\alpha_* & \xrightarrow{\sim} & \alpha'_*(\beta')^{-1}\alpha^{-1}\alpha_* \end{array}$$

The claim is that this diagram commutes, and it can be proven by filling out the diagram using the naturality of the units, counits, and the triangle identities.

The push-pull map is also called the **change of basis map**.

Definition 1.6.7

Let $\mathcal{F}$ be a sheaf of Abelian groups on X , and $s \in \mathcal{F}(X)$ a global section. Define its **support** to be

$$\text{supp } s = \{p \in X \mid s_p \neq 0 \text{ in } \mathcal{F}_p\}.$$

Proposition 1.6.8

The support of a global section $\text{supp } s$ is a closed subset of X .

Proof

Note that $p \notin \text{supp } s$ iff $s_p = 0$ in $\mathcal{F}_p$ iff there is a neighborhood $p \in U$ such that $s|_U = 0$ in $\mathcal{F}(U)$. But then for all $q \in U$ one has $s_q = 0$ for this reason, so $U \subseteq (\text{supp } s)^c$; so $\text{supp } s$ is closed. ■

Note:

The locus where a continuous function is nonzero is open, i.e. if f is a continuous function to an Abelian group then $\{p \in X \mid f(p) \neq 0\}$ is open. In contrast the previous proposition shows that the locus where the germ of a continuous function (i.e. the stalk of a sheaf of continuous functions) is nonzero is closed.

This is because the latter is the set of all points where s is nonzero *near* (i.e. in a neighborhood around) p , while the former is the set of all points where f is nonzero *at* p .

In the following sheaves are valued in a category with a terminal object (so not **Ring** for example).

Definition 1.6.9

The **support** of a sheaf $\mathcal{F} \in \mathcal{C}_X$ is the set $\text{supp } \mathcal{F}$ of $p \in X$ for which $\mathcal{F}_p$ is non-terminal in $\mathcal{C}$.

If $\mathcal{F}$ is a sheaf of groups, then

$$\text{supp } \mathcal{F} = \{p \in X \mid \mathcal{F}_p \neq 0\} = \bigcup_{s \in \Gamma(\mathcal{F})} \text{supp } s.$$

Proposition 1.6.10

Let $Z \subseteq X$ be a closed subset, and denote by $\iota: Z \hookrightarrow X$ the inclusion.

- (1) If $\mathcal{F}$ is a sheaf on Z then for $q \in X$,

$$(\iota_* \mathcal{F})_q = \begin{cases} \mathcal{F}_q & q \in Z \\ * & q \notin Z \end{cases}$$

where $*$ $\in \mathcal{C}$ is the terminal object.

- (2) If $\mathcal{F}$ is a sheaf on X with $\text{supp } \mathcal{F} \subseteq Z$ then the unit $\mathcal{F} \rightarrow \iota_* \iota^{-1} \mathcal{F}$ is an isomorphism.

Proof

For (1) note that

$$(\iota_* \mathcal{F})_q = \text{colim}_{q \in U} \mathcal{F}(U \cap Z).$$

If $q \in Z$ then this is the colimit over all neighborhoods of q in Z , so isomorphic to $\mathcal{F}_q$. Otherwise there is a neighborhood of q disjoint from Z as it is closed, and $\mathcal{F}(\emptyset) = *$.

For (2), let $p \in X$. If $p \in Z$ then by (1) and corollary 1.6.4

$$(\iota_* \iota^{-1} \mathcal{F})_p \cong (\iota^{-1} \mathcal{F})_p \cong \mathcal{F}_p.$$

This isomorphism is the stalk of the unit. And if $p \notin Z$ then $p \notin \text{supp } \mathcal{F}$ so that $\mathcal{F}_p$ is terminal and so is $(\iota_* \iota^{-1} \mathcal{F})_p$ by (1).

Thus the stalks of the unit are all isomorphisms, and thus the unit itself is an isomorphism. ■

2 Schemes

2.1 Affine schemes

In this section we briefly review the definition of an affine scheme.

Definition 2.1.1

Let A be a commutative ring; its **spectrum** is the topological space whose underlying set is the set of all

prime ideals of A . We denote it by $\text{Spec}(A)$, and for a prime ideal $\mathfrak{p} \leq A$ we denote the element of $\text{Spec}(A)$ by $[\mathfrak{p}] \in \text{Spec}(A)$.

For a subset $S \subseteq A$ we define its **vanishing set** to be

$$V(S) = \{[\mathfrak{p}] \in \text{Spec}(A) \mid S \subseteq \mathfrak{p}\}.$$

The collection of all such vanishing sets, $\{V(S)\}_{S \subseteq A}$, forms a topology of closed sets on $\text{Spec}(A)$. This topology is called the **Zariski topology**.

Notice that for a subset $S \subseteq A$, $V(S) = V((S))$ where $(S) \leq A$ is the ideal generated by S . Furthermore,

$$V(\emptyset) = V(0) = \text{Spec}(A), \quad V(A) = \emptyset, \quad \bigcap_{i \in I} V(S_i) = V\left(\bigcup_{i \in I} S_i\right),$$

and that for two ideals $I, J \leq A$ we have that $V(I) \cup V(J) = V(IJ)$. It follows that the Zariski topology is indeed a topology. Notice that V is order-reversing: if $S_1 \subseteq S_2$ then $V(S_2) \subseteq V(S_1)$.

Recall that if $\varphi: A \rightarrow B$ is a ring morphism then it pulls back prime ideals: if $\mathfrak{q} \leq B$ is prime, then so is $\varphi^{-1}\mathfrak{q} \leq A$. Thus Spec is functorial:

$$\text{Spec}: \text{Ring}^{\text{op}} \rightarrow \text{Top}, \quad \varphi \mapsto \varphi^*$$

where $\varphi^*: \text{Spec}(B) \rightarrow \text{Spec}(A)$ is $[\mathfrak{q}] \mapsto [\varphi^{-1}\mathfrak{q}]$. This is continuous: for $I \leq A$

$$(\varphi^*)^{-1}V_A(I) = \{[\mathfrak{q}] \in \text{Spec}(B) \mid I \subseteq \varphi^{-1}\mathfrak{q}\} = \{[\mathfrak{q}] \in \text{Spec}(B) \mid \varphi(I) \subseteq \mathfrak{q}\} = V_B(\varphi(I)).$$

If $I \leq A$ is an ideal then its **radical** is the ideal

$$\sqrt{I} = \{a \in A \mid a^n \in I \text{ for some } n \geq 0\} \leq A.$$

It is easy to see that $V(I) = V(\sqrt{I})$: if $I \leq \mathfrak{p}$ then if $f \in \sqrt{I}$ then $f^n \in I \leq \mathfrak{p}$ for some n and since $\mathfrak{p}$ is prime, $f \in \mathfrak{p}$. Furthermore it can be seen that $\sqrt{I}$ is the intersection of all prime ideals containing I :

$$\sqrt{I} = \bigcap_{I \leq \mathfrak{p}} \mathfrak{p} = \bigcap V(I).$$

Clearly the left-hand side is contained in the right. Conversely if $f \notin \sqrt{I}$ then $\sqrt{I}$ and $\{f^n\}_{n \geq 0}$ do not intersect, and so $(\sqrt{I})_f \leq A_f$ is a proper ideal. Thus it is contained in a prime ideal $(\sqrt{I})_f \leq \mathfrak{p}_f \leq A_f$ associated to a prime ideal $\mathfrak{p} \leq A$, so that $\sqrt{I} \leq \mathfrak{p}$. But $f \notin \mathfrak{p}$, so $f \notin \bigcap V(I)$.

We also define the **nilradical** of A be the radical of the zero ideal:

$$\text{nil}(A) = \sqrt{0} = \{a \in A \mid a^n = 0 \text{ for some } n \geq 0\}.$$

Thus we have that

$$\text{nil}(A) = \bigcap_{[\mathfrak{p}] \in \text{Spec}(A)} \mathfrak{p}.$$

We can also form a base for the Zariski topology of $\text{Spec}(A)$. For $S \subseteq A$ let

$$D(S) = V(S)^c = \{[\mathfrak{p}] \in \text{Spec}(A) \mid S \not\subseteq \mathfrak{p}\}$$

form the open sets of the Zariski topology. For $f \in A$ let

$$D(f) = D\{f\} = \{[\mathfrak{p}] \in \text{Spec}(A) \mid f \notin \mathfrak{p}\}.$$

It is clear that

$$D(S) = \bigcup_{f \in S} D(f)$$

so that the $D(f)$ form a base of open sets for the topology.

We can map ideals of A to closed subsets of $\text{Spec}(A)$, we can also map subsets of $\text{Spec}(A)$ to ideals of A . For $X \subseteq \text{Spec}(A)$ we define the ideal

$$I(X) = \bigcap_{[\mathfrak{p}] \in X} \mathfrak{p} \leq A.$$

This is also clearly order-reversing: if $X_1 \subseteq X_2$ then $I(X_2) \subseteq I(X_1)$.

Notice that for an ideal $I \leq A$,

$$I(V(I)) = \bigcap_{I \leq \mathfrak{p}} \mathfrak{p} = \sqrt{I}.$$

And conversely for a subset $X \subseteq \text{Spec}(A)$,

$$V(I(X)) = \overline{X},$$

the closure of X in the Zariski topology. Indeed: if $X \subseteq V(I)$ then for $[\mathfrak{p}] \in X$ we have $I \leq \mathfrak{p}$ so that $I \leq \bigcap_{[\mathfrak{p}] \in X} \mathfrak{p} = I(X)$. Thus $V(I(X)) \subseteq V(I)$, so that $V(I(X))$ is the smallest closed subset containing X ; i.e. $\overline{X}$. Thus V and I form order-reversing bijections

$$\{\text{radical ideals of } A\} \xrightleftharpoons[I(\bullet)]{V(\bullet)} \{\text{closed subsets of } \text{Spec}(A)\}.$$

It follows that $V(I) \subseteq V(J)$ iff $\sqrt{J} \leq \sqrt{I}$: applying $I(\bullet)$ gives us the left-to-right direction, and $V(\sqrt{I}) = V(I)$ gives the right-to-left direction. In particular $V(I) = V(J)$ iff $\sqrt{I} = \sqrt{J}$. This carries over to D as well obviously: $D(I) \subseteq D(J)$ iff $\sqrt{I} \leq \sqrt{J}$. In particular $D(f) \subseteq D(g)$ iff $\sqrt{f} \leq \sqrt{g}$ iff $f^n \in (g)$ for some $n \geq 0$, or equivalently g is invertible in A_f (g is invertible in A_f iff $(g)_f$ is unital, iff (g) intersects the localizing set, iff $f^n \in (g)$ for some $n \geq 0$).

Because $V(0) = \text{Spec}(A)$, $V(I) = \text{Spec}(A)$ iff $\sqrt{I} = \text{nil}(A)$ iff $I \subseteq \text{nil}(A)$. In particular $D(f) = \emptyset$ iff $f \in \text{nil}(A)$. On the other hand, $V(I) = \emptyset$ for an ideal I iff $I = A$, since every proper ideal is contained in a prime ideal. Thus $D(f) = \text{Spec}(A)$ iff f is a unit.

We now discuss some topological results regarding $\text{Spec}(A)$.

Definition 2.1.2

A topological space X is said to be

- (1) **quasicompact** if every open cover of X has a finite subcover. Explicitly, if $X = \bigcup_{i \in I} U_i$ for $U_i \subseteq X$, then there are $i_1, \dots, i_n \in I$ such that $X = \bigcup_{j=1}^n U_{i_j}$.
- (2) **connected** if X cannot be written as the disjoint union of two nonempty open sets. Equivalently, X has no clopen subsets (subsets which are simultaneously open and closed) other than X and $\emptyset$. Otherwise X is **disconnected**.
- (3) **irreducible** if X cannot be written as the union of two proper closed subsets. That is if $X = Y \cup Z$ for Y, Z closed then either $X = Y$ or $X = Z$. Equivalently the intersection of any two nonempty open subsets is nonempty.

Note:

The term “quasicompact” may be odd to you, as this is sometimes taken to be the definition of a “compact” space. For geometers, oftentimes a compact space is a space which is both quasicompact and Hausdorff.

Remark 2.1.3

A space X is quasicompact iff every collection of closed sets with the finite intersection property has nonempty intersection. That is, if $\{Z_i\}_{i \in I}$ is a collection of closed subsets of X such that $\bigcap_{i \in I_0} Z_i$ is nonempty for all finite $I_0 \subseteq I$, then $\bigcap_{i \in I} Z_i \neq \emptyset$.

Remark 2.1.4

It is immediate that a space X is irreducible iff all nonempty open subsets of X are dense.

It is true that $\text{Spec}(A)$ is always quasicompact. It is sufficient to consider basic closed sets. If $\{V(f_i)\}_{i \in I}$ has the finite intersection property, then

$$\bigcap_{i \in I} V(f_i) = V(\{f_i\}_{i \in I})$$

and if this is empty, then $(f_i \mid i \in I) = A$. So there are $i_1, \dots, i_n \in I$ such that $1 \in (f_{i_1}, \dots, f_{i_n})$ so that

$$\emptyset = V(f_{i_1}, \dots, f_{i_n}) = \bigcap_{j=1}^n V(f_{i_j}),$$

contradicting the finite intersection property.

Definition 2.1.5

An **idempotent** in a ring A is an element $f \in A$ such that $f^2 = f$. A **nontrivial idempotent** is an idempotent which is not 0 or 1. A ring A is **connected** if it has no nontrivial idempotents; otherwise it is **disconnected**.

Proposition 2.1.6

Let A be a ring. Then the following conditions are equivalent:

- (1) A is disconnected;
- (2) A is isomorphic to the product of two non-trivial rings;
- (3) $\text{Spec}(A)$ is disconnected.

In particular $\text{Spec}(A \times B) \cong \text{Spec}(A) \amalg \text{Spec}(B)$.

Proof

Suppose (1), then if $f \in A$ is a nontrivial idempotent let $g = 1 - f$; this is also a nontrivial idempotent. The ideal $A^f = (f)$ has a ring structure whose addition and multiplication is inherited from A , its identity is f . We define $\varphi: A \rightarrow A^f \times A^g$ by $a \mapsto (fa, ga)$. This is injective because $fa + ga = a$, and it is surjective because $fg = 0$ so $\varphi(fa + gb) = (fa, gb)$. Thus $A \cong A^f \times A^g$, and these are nontrivial because $f, g \neq 0$; so (1) implies (2).

Conversely if $A \cong B \times C$ then $(1, 0), (0, 1) \in A$ are nontrivial idempotents so A is disconnected. Thus (2) implies (1).

Now again suppose (2), so consider the ring $A \times B$. Then notice that $\text{Spec}(A \times B) = D(1, 0) \sqcup D(0, 1)$. Indeed:

$$D(1, 0) \cup D(0, 1) = D((1, 0), (0, 1)) = D(1) = \text{Spec}(A \times B), \quad D(1, 0) \cap D(0, 1) = D((1, 0) \cdot (0, 1)) = D(0) = \emptyset.$$

Thus (2) implies (3).

Notice that the projection maps $A \times B \rightarrow A, B$ induce $\text{Spec}(A) \amalg \text{Spec}(B) \rightarrow \text{Spec}(A \times B)$. It is easy to see that this maps $\text{Spec}(A)$ to $D(1, 0)$ and $\text{Spec}(B)$ to $D(0, 1)$, showing that this is an isomorphism.

Now assume (3), so $\text{Spec}(A) = V(I) \sqcup V(J)$ where I, J are nonzero (so $V(I), V(J)$ are proper subsets). This means that $V(I + J) = \emptyset$ so that $I + J = A$ and $V(IJ) = \text{Spec}(A)$ so that $IJ = 0$. Thus by the Chinese remainder theorem, $A \rightarrow A/I \times A/J$ is an isomorphism, proving (2). ■

Proposition 2.1.7

$\text{Spec}(A)$ is irreducible iff $\text{nil}(A)$ is prime.

Proof

If $\text{nil}(A)$ is not prime then there are $f, g \notin \text{nil}(A)$ such that $fg \in \text{nil}(A)$. So $V(f), V(g) \neq \text{Spec}(A)$ and $\text{Spec}(A) = V(fg) = V(f) \cup V(g)$; thus $\text{Spec}(A)$ is reducible as desired.

Conversely if $\text{Spec}(A)$ is reducible then suppose $\text{Spec}(A) = V(I) \cup V(J)$ for proper subsets, so I, J are not subsets of $\text{nil}(A)$. But then $V(IJ) = \text{Spec}(A)$ so $IJ \leq \text{nil}(A)$, so $\text{nil}(A)$ is not prime. ■

Corollary 2.1.8

If A is a domain then $\text{Spec}(A)$ is irreducible.

Note:

While it is true that $\text{Spec} \prod_{i=1}^n A_i \cong \prod_{i=1}^n \text{Spec} A_i$ for finite products, this does not hold for infinite products. There is a canonical map $\prod_{i \in I} \text{Spec} A_i \rightarrow \text{Spec} \prod_{i \in I} A_i$; and it is easy to see that this map is an embedding. But if all the A_i are nonzero and I is infinite, the left hand side is not compact (because $\{\text{Spec} A_i\}_{i \in I}$ does not have a finite subcover) and thus it cannot be an isomorphism.

Definition 2.1.9

Let X be a topological space. A **closed point** of X is a point $x \in X$ such that $\overline{\{x\}} = \{x\}$.

In a Hausdorff space, all points are closed. For $\text{Spec} A$, we know that $\overline{[\mathfrak{p}]} = V(\text{I}([\mathfrak{p}])) = V(\mathfrak{p})$. Thus $[\mathfrak{p}]$ is a closed point iff $\mathfrak{p}$ is not contained in any prime ideal other than itself; equivalently it is a maximal ideal. That is: $[\mathfrak{p}] \in \text{Spec} A$ is a closed point iff $\mathfrak{p} \leq A$ is maximal.

Definition 2.1.10

A subset of a topological space $Z \subseteq X$ is an **irreducible component** if it is irreducible and maximally so: it is not properly contained in an irreducible set.

Proposition 2.1.11

A subset $Z \subseteq X$ is irreducible iff $\overline{Z} \subseteq X$ is.

Proof

If Z is irreducible and $\overline{Z} = Y \cup W$ are closed, then $Z \subseteq Y \cup W$ so that $Z \subseteq Y$ or $Z \subseteq W$. Then $\overline{Z} = Y$ or W because they are closed in $\overline{Z}$ and thus in X . Conversely if $\overline{Z}$ is irreducible then if $Z = Y \cup W$, then $\overline{Z} = \overline{Y} \cup \overline{W}$ so $\overline{Z} = \overline{Y}$ or $\overline{W}$. Suppose the former, so $Z = \overline{Y} \cap Z = Y$ since $Y \subseteq Z$ is closed. ■

Proposition 2.1.12

Every point $p \in X$ is contained in a (not necessarily unique) irreducible component, and all irreducible components of X are closed.

Proof

The first claim is due to Zorn. Clearly $\{p\}$ is irreducible, so we can form the nonempty poset of irreducible sets containing p . For a chain of such irreducible sets $\{Z_i\}_{i \in I}$, we have that $Z = \bigcup_{i \in I} Z_i$ is irreducible. Indeed if $Z = Y \cup W$ are closed then suppose $Y \subsetneq Z$ then there is an $f \in Z_i$ such that $f \notin Y$. Because Z_i is irreducible we must have $Z_i \subseteq W$. Then for all $j \geq i$ we have that $Z_j \subseteq W$ for the same reason; then $Z = \bigcup_{j \geq i} Z_j = W$ so that Z is irreducible as desired.

The second claim follows from the previous proposition: if Z is an irreducible component then $\overline{Z}$ is irreducible as well and since $Z \subseteq \overline{Z}$ by maximality $Z = \overline{Z}$. ■

Definition 2.1.13

Let $x, y \in X$ be points in a topological space such that $x \in \overline{\{y\}}$. Then x is called a **specialization** of y , and y a **generalization** of x .

A point $p \in X$ is a **generic point** if $\overline{\{p\}} = X$; i.e. if it is a generalization of all the points of X .

In $\text{Spec } A$ because $\overline{[\mathfrak{p}]} = V(\mathfrak{p})$, $[\mathfrak{p}]$ is a generalization of $[\mathfrak{q}]$ iff $[\mathfrak{q}] \in V(\mathfrak{p})$ iff $\mathfrak{p} \leq \mathfrak{q}$. And $[\mathfrak{p}]$ is a generic point iff $V(\mathfrak{p}) = \text{Spec}(A)$ iff $\mathfrak{p} = \text{nil } A$, so that $\text{Spec } A$ has a generic point iff $\text{nil } A$ is prime iff $\text{Spec } A$ is irreducible.

Definition 2.1.14

A poset Σ has the **descending chain condition (DCC)** if any of the following two equivalent conditions are satisfied:

- (1) every subset $\Sigma' \subseteq \Sigma$ has a minimal element;
- (2) every chain of elements $p_1 \geq p_2 \geq p_3 \geq \dots$ eventually stabilizes: $p_n = p_{n+1} = p_{n+2} = \dots$ for some n onward.

Dually Σ has the **ascending chain condition (ACC)** if Σ^{op} has the DCC.

The equivalence of the conditions is clear: (1) implies (2) by defining $\Sigma' = \{p_n\}_n$ and taking the minimum. Conversely if $\Sigma' \subseteq \Sigma$ has no minimal element, then we can form an infinite decreasing chain.

Definition 2.1.15

A topological space X is **Noetherian** if its poset of closed subsets has the DCC (equivalently its poset of open subsets has the ACC).

There is a notion of **Noetherian induction**: if Σ has DCC and we have some property P (which one can view as a function $\Sigma \rightarrow \{0, 1\}$) such that $P(x)$ holds for all minimal elements of Σ , and for $y \in \Sigma$ non-minimal if $P(x)$ holds for all $x < y$ then $P(y)$. Then $P(x)$ holds for all $x \in \Sigma$. This is easy to prove: take Σ' to be the set of all $x \in \Sigma$ for which $P(x)$ does not hold.

Using Noetherian induction one can prove the following.

Proposition 2.1.16

If X is a Noetherian topological space, then every nonempty closed subset $Z \subseteq X$ can be uniquely expressed as a finite union $Z = Z_1 \cup \cdots \cup Z_n$ where the Z_i are all irreducible closed subsets, none containing any other.

Remark 2.1.17

It is clear that if A is a Noetherian ring (its poset of ideals has ACC) then $\text{Spec } A$ is a Noetherian space. The converse is not true: if $\text{Spec } A$ is a Noetherian space then the poset of A 's *radical* ideals has ACC.

2.2 The structure sheaf on an affine scheme

Definition 2.2.1

Let A be a ring, then we define a sheaf of rings $(\text{Spec } A, \mathcal{O}_{\text{Spec } A})$ as follows. For a basic open set $D(f)$, define $\mathcal{O}_{\text{Spec } A}(D(f))$ to be the localization of A at all functions which do not vanish outside of $V(f)$, i.e. all $g \in A$ such that $V(g) \subseteq V(f)$.

For $f \in A$ let $S = \{g \in A \mid V(g) \subseteq V(f)\} = \{g \in A \mid D(f) \subseteq D(g)\}$. Recall that $V(g) \subseteq V(f)$ iff $f^n \in (g)$ iff $g \in A_f$ is invertible. Thus S is the saturation of $\{f^n\}_{n \geq 0}$, so that $S^{-1}A = \mathcal{O}_{\text{Spec } A}(D(f))$ and A_f are naturally isomorphic.

If $D(f') \subseteq D(f)$ then there is an obvious map $\mathcal{O}_{\text{Spec } A}(D(f)) \rightarrow \mathcal{O}_{\text{Spec } A}(D(f'))$ as the right ring is a localization of the left. Taking these maps as the restriction maps, this makes $\mathcal{O}_{\text{Spec } A}$ into a presheaf of rings (or A -algebras). To show that $\mathcal{O}_{\text{Spec } A}$ is a sheaf, we will make use of the following lemmas.

Lemma 2.2.2

Let A be a ring and $f_1, \dots, f_n$ generate the unit ideal. Then the following sequence of A -module maps is exact

$$0 \rightarrow A \xrightarrow{\varphi} \prod_i A_{f_i} \xrightarrow{\psi} \prod_{i,j} A_{f_i f_j},$$

where $\varphi(a) = (a)_i$ and $\psi(a_i) = (a_i - a_j)_{i,j}$. (We write $a \in A_f$ for the localization $a = a/1$.)

Proof

First we show that φ is injective: if $a = 0$ in all A_{f_i} then there are $m_1, \dots, m_n$ such that $f_i^{m_i} a = 0$ for all $i = 1, \dots, n$. But because $f_i \in \sqrt{(f_j^{m_j})_j}$ for all i and $(f_1, \dots, f_n) = A$, we see that $(f_1^{m_1}, \dots, f_n^{m_n}) = A$ as well. Thus there are $g_1, \dots, g_n \in A$ such that $g_1 f_1^{m_1} + \cdots + g_n f_n^{m_n} = 1$; multiplying by a shows that $a = 0$. Next we show that $\psi \circ \varphi = 0$. Indeed:

$$\psi \circ \varphi(a) = \psi(a)_i = (a - a)_{i,j} = 0,$$

so that $\text{im } \varphi \subseteq \ker \psi$. Now suppose that $(a_i/f_i^{m_i})_i \in \ker \psi$, then by multiplying both the numerator and denominator by f_i^s , we can assume that all the m_i 's are equal to some m . Thus

$$0 = \psi(a_i/f_i^m)_i = (a_i/f_i^m - a_j/f_j^m)_{i,j}$$

so that there is an M where $(f_i f_j)^M (f_j^m a_i - f_i^m a_j) = 0$ for all i, j .

As before, the f_i^{M+m} generate the unit ideal, so there are $b_i \in A$ such that $\sum_i b_i f_i^{M+m} = 1$. Then define $a = \sum_i b_i f_i^M a_i$ so that

$$f_j^{M+m} a = \sum_i b_i (f_i f_j)^M f_j^m a_i = \sum_i b_i (f_i f_j)^M f_i^m a_j = f_j^M \sum_i b_i f_i^{M+m} a_j = f_j^M a_j.$$

Thus in A_f , $a = a_j/f_j^m$, so that $\varphi(a) = (a_i/f_i^m)_i$ as desired. $\blacksquare$

We will say that a presheaf on X **has the sheaf property at U** for an open $U \subseteq X$ if the sheaf axiom holds for all open covers of U . So a sheaf is a presheaf which has the sheaf property at every open subset of X . Furthermore we can say that a presheaf has the sheaf property at U **relative to $\{U_i\}_i$** for a specified open cover $\{U_i\}_i$ of U if the sheaf condition holds specifically for $\{U_i\}_i$.

Lemma 2.2.3

If X is a quasicompact space, then a presheaf $\mathcal{F}$ has the sheaf property at X iff it has the sheaf property at X relative to all finite covers of X .

Proof

One direction is clear. Conversely let $\mathcal{F}$ be a presheaf with the sheaf property at X relative to all finite covers of X , and let $\mathcal{U} = \{U_i\}_{i \in I}$ be an open cover of X . Then there is a finite subset $I_0 \subseteq I$ such that $\mathcal{U}_0 = \{U_i\}_{i \in I_0}$ covers X . To show that $\mathcal{F}(U) \Rightarrow \mathcal{F}|_{\mathcal{U}}$ is a limiting cone (see remark 1.1.6), we note that a cone over $\mathcal{F}|_{\mathcal{U}}$ induces a cone over $\mathcal{F}|_{\mathcal{U}_0}$. Because $\mathcal{F}(U) \Rightarrow \mathcal{F}|_{\mathcal{U}_0}$ is limiting, any morphism of cones $(C \Rightarrow \mathcal{F}|_{\mathcal{U}}) \rightarrow (\mathcal{F}(U) \Rightarrow \mathcal{F}|_{\mathcal{U}})$ must be unique.

To show existence, it is sufficient to show that a morphism of cones $(C \Rightarrow \mathcal{F}|_{\mathcal{U}_0}) \rightarrow (\mathcal{F}(U) \Rightarrow \mathcal{F}|_{\mathcal{U}_0})$ is also a morphism of cones over $\mathcal{F}|_{\mathcal{U}}$. If not, then there is a finite diagram which doesn't commute: there is a $V \in \mathcal{U}$ such that $C \rightarrow \mathcal{F}(V)$ is not equal to $C \rightarrow \mathcal{F}(U) \rightarrow \mathcal{F}(V)$. Adding V to $\mathcal{U}_0$ yields a finite cover of U , and since $\mathcal{F}$ has the sheaf property relative to *all* finite covers of X , this is a contradiction. $\blacksquare$

Theorem 2.2.4

$\mathcal{O}_{\text{Spec } A}$ is a sheaf on the distinguished base, thus determines a unique sheaf on the topological space $\text{Spec } A$.

Proof

We must show that if $D(f) = \bigcup_{i \in I} D(f_i)$ then the following sequence is exact

$$0 \rightarrow A_f \xrightarrow{\varphi} \prod_i A_{f_i} \xrightarrow{\psi} \prod_{i,j} A_{f_i f_j}$$

where φ, ψ are defined as above (recall that $D(f) \cap D(g) = D(fg)$). For a finite I , this follows from the lemma before the previous with A_f in place of A . For general I this follows from the above lemma, along with the fact that $D(f) \cong \text{Spec}(A_f)$ is quasicompact. $\blacksquare$

Note:

It is *not* the case that $\mathcal{O}_{\text{Spec } A}(U)$ is the localization of A at all $f \in A$ not vanishing outside of U .

For a ring A , we can embed Mod_A fully faithfully into $\text{Mod}_{\mathcal{O}_{\text{Spec } A}}$. For an A -module M , let $\widetilde{M} \in \text{Mod}_{\mathcal{O}_{\text{Spec } A}}$ be the presheaf where $\widetilde{M}(D(f))$ is the localization of M at all $g \in A$ which don't vanish outside of $V(f)$. For the same reason as before, $\widetilde{M}(D(f)) \cong M_f$. The restriction maps are similar to before, and the proof that $\widetilde{M}$ is a B -sheaf, and thus determines a sheaf, is very similar to above.

In particular, $\widetilde{A}$ is just $\text{Spec } A$.

Lemma 2.2.5

Let A be a ring and $S \subseteq A$ a multiplicative set. We can view S as a preorder whose objects are the $f \in S$, and there is an arrow $f \rightarrow g$ iff $D(f) \subseteq D(g)$. Then for an A -module M , $f \mapsto M_f$ forms a functor $S^{\text{op}} \rightarrow \text{Mod}_{S^{-1}A}$, and

$$S^{-1}M \cong \text{colim}_{f \in S^{\text{op}}} M_f.$$

Proof

It is sufficient to prove this for $M = A$, as S is a cofiltered category ($fg \rightarrow f, g$) so the colimit is filtered. Because tensor products commute with filtered colimits,

$$\text{colim } M_f \cong \text{colim}(A_f \otimes M) \cong (\text{colim } A_f) \otimes M \cong S^{-1}A \otimes M \cong S^{-1}M.$$

The following proof works just as well for M , but is clearer with A .

Let $\varphi: A \rightarrow B$ be a ring map such that $\varphi(S) \subseteq B^\times$. Then $\varphi(f) \in B^\times$ so that this induces unique maps $\psi_f: A_f \rightarrow B$ for all $f \in S$ such that the following diagrams commute:

$$\begin{array}{ccc} A & \xrightarrow{\varphi} & B \\ & \searrow \iota_f & \uparrow \psi_f \\ & & A_f \end{array}$$

By uniqueness this forms a cone $(f \mapsto A_f) \Rightarrow B$, and thus a unique map $\text{colim } A_f \rightarrow B$. So we have a commutative diagram

$$\begin{array}{ccc} A & \xrightarrow{\varphi} & B \\ & \searrow & \uparrow \\ & & \text{colim } A_f \end{array}$$

Because $1 \in S$ and $A = A_1$, this gives rise to the desired map. This also shows uniqueness of the map: a map $\text{colim } A_f \rightarrow B$ commuting with φ gives rise to maps $A_f \rightarrow B$ commuting with $A \rightarrow A_f$ (again because $A = A_1$), and by the universal property of A_f , they are unique. ■

Corollary 2.2.6

Let $M \in \text{Mod}_A$ be an A -module and $\widetilde{M} \in \text{Mod}_{\mathcal{O}_{\text{Spec } A}}$ be the induced $\mathcal{O}_{\text{Spec } A}$ -module. Then for $[\mathfrak{p}] \in \text{Spec } A$ there is a natural isomorphism between the stalk of $\widetilde{M}$ at $[\mathfrak{p}]$ and the localization $M_{\mathfrak{p}}$:

$$\widetilde{M}_{[\mathfrak{p}]} \cong M_{\mathfrak{p}}.$$

In particular

$$(\text{Spec } A)_{[\mathfrak{p}]} \cong A_{\mathfrak{p}}.$$

Proof

The lefthand side is

$$\widetilde{M}_{[\mathfrak{p}]} = \text{colim}_{[\mathfrak{p}] \in D(f)} \widetilde{M}(D(f)) = \text{colim}_{f \notin \mathfrak{p}} M_f,$$

which is $(A - \mathfrak{p})^{-1}M = M_{\mathfrak{p}}$ by the previous lemma. ■

Proposition 2.2.7

Let M be an A -module. Then the natural map

$$M \hookrightarrow \prod_{\mathfrak{p} \in \operatorname{Spec} A} M_{\mathfrak{p}}$$

is injective.

Proof

This map corresponds to the map

$$M = \widetilde{M}(\operatorname{Spec} A) \hookrightarrow \prod_{\mathfrak{p} \in \operatorname{Spec} A} \widetilde{M}_{\mathfrak{p}} \cong \prod_{\mathfrak{p} \in \operatorname{Spec} A} M_{\mathfrak{p}}$$

which is an injection by proposition 1.3.1. ■

Note:

Alternatively we could prove this directly: if $m \in M$ is equal to zero in $M_{\mathfrak{p}}$ for all $\mathfrak{p} \in \operatorname{Spec} A$. Define the ideal

$$I = \operatorname{ann}_A(m) = \{a \in A \mid am = 0\} \leq A.$$

If m were nonzero, then I is a proper ideal and thus contained in a prime ideal $\mathfrak{p}$. For $a \notin \mathfrak{p}$ then $a \notin I$ so $am \neq 0$, and thus $m \neq 0$ in $M_{\mathfrak{p}}$ a contradiction.

Note that $M \mapsto \widetilde{M}$ is functorial. Indeed, an A -module map $\varphi: M \rightarrow N$ defines maps

$$\varphi_f: \widetilde{M}(\operatorname{D} f) = M_f \rightarrow N_f = \widetilde{N}(\operatorname{D} f)$$

which assemble into a morphism $\widetilde{\varphi}: \widetilde{M} \rightarrow \widetilde{N}$.

Note that this functor is fully faithful. It is faithful because $\widetilde{\varphi}_{\operatorname{D} 1} = \varphi$. And it is full because if $\psi: \widetilde{M} \rightarrow \widetilde{N}$ is a morphism, then let $\varphi = \psi_{\operatorname{D} 1}$, and then $\psi = \widetilde{\varphi}$. Indeed, by naturality we must have commutative squares for all $f \in A$:

$$\begin{array}{ccc} M & \xrightarrow{\quad \varphi \quad} & N \\ \downarrow & & \downarrow \\ M_f & \xrightarrow{\quad \psi_{\operatorname{D} f} \quad} & N_f \end{array}$$

By the universal property of localization we must have $\psi_{\operatorname{D} f} = \varphi_f$ as desired. Thus $M \mapsto \widetilde{M}$ is indeed a fully faithful embedding $\mathbf{Mod}_A \rightarrow \mathbf{Mod}_{\operatorname{Spec} A}$.

Definition 2.2.8

Let M be an A -module and $m \in M$. Define its **support** to be

$$\operatorname{supp} m = \{[\mathfrak{p}] \in \operatorname{Spec} A \mid m_{\mathfrak{p}} \neq 0\},$$

which is the support of m viewed as a global section of $\widetilde{M}$. We similarly define

$$\operatorname{supp} M = \operatorname{supp} \widetilde{M} = \{[\mathfrak{p}] \in \operatorname{Spec} A \mid M_{\mathfrak{p}} \neq 0\}.$$

2.3 Schemes

Note that when $\pi: X \rightarrow Y$ is a homeomorphism, there is an isomorphism of categories $\pi: \text{open}(Y) \cong \text{open}(X)$ mapping $V \mapsto \pi^{-1}V$. This lifts to an isomorphism of categories $\pi_*: \mathcal{C}_X^{\text{pre}} \cong \mathcal{C}_Y^{\text{pre}}$ mapping a presheaf $\mathcal{F} \in \mathcal{C}_X^{\text{pre}}$ to precomposition with π : $\pi_*\mathcal{F}(V) = \mathcal{F}(\pi^{-1}V)$. This is the usual pushforward, and it then induces an isomorphism $\mathcal{C}_X \cong \mathcal{C}_Y$.

Because π^{-1} is its left adjoint, which is unique up to isomorphism, this means that π^{-1}, π_* form an adjoint equivalence. In particular, the adjunction preserves categorical properties of morphisms, e.g. a morphism $\mathcal{F} \rightarrow \pi_*\mathcal{G}$ is an isomorphism iff $\pi^{-1}\mathcal{F} \rightarrow \mathcal{G}$ is.

Definition 2.3.1

Let $\mathcal{C}$ be a category (with enough limits and colimits). A **$\mathcal{C}$ -space** is a pair $(X, \mathcal{O}_X)$ where X is a topological space and $\mathcal{O}_X$ is a $\mathcal{C}$ -valued sheaf on X . A morphism of $\mathcal{C}$ -spaces $(X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ is a continuous map $\pi: X \rightarrow Y$ along with a map of sheaves $\varphi: \mathcal{O}_Y \rightarrow \pi_*\mathcal{O}_X$. Composition is given in the obvious manner: if $X \xrightarrow{\pi} Y \xrightarrow{\tau} Z$ are continuous, and $\varphi: \mathcal{O}_Y \rightarrow \pi_*\mathcal{O}_X$ and $\psi: \mathcal{O}_Z \rightarrow \tau_*\mathcal{O}_Y$ are morphisms, then define

$$(\tau, \psi) \circ (\pi, \varphi) = \mathcal{O}_Z \xrightarrow{\psi} \tau_*\mathcal{O}_Y \xrightarrow{\tau_*\varphi} (\tau \circ \pi)_*\mathcal{O}_X.$$

We denote the category of $\mathcal{C}$ -spaces by $\text{Space}_{\mathcal{C}}$.

Note that composition is indeed associative. If we have maps

$$(X, \mathcal{O}_X) \xrightarrow{(\pi, \varphi)} (Y, \mathcal{O}_Y) \xrightarrow{(\tau, \psi)} (Z, \mathcal{O}_Z) \xrightarrow{(\sigma, \chi)} (W, \mathcal{O}_W)$$

then we have compositions

$$\begin{aligned} ((\sigma, \chi) \circ (\tau, \psi)) \circ (\pi, \varphi) &= (\sigma \circ \tau, \sigma_*\psi \circ \chi) \circ (\pi, \varphi) = (\sigma \circ \tau \circ \pi, (\sigma \circ \tau)_*\varphi \circ \sigma_*\psi \circ \chi) \\ (\sigma, \chi) \circ ((\tau, \psi) \circ (\pi, \varphi)) &= (\sigma, \chi) \circ (\tau \circ \pi, \tau_*\varphi \circ \psi) = (\sigma \circ \tau \circ \pi, \sigma_*(\tau_*\varphi \circ \psi) \circ \chi) \end{aligned}$$

which are equal.

Note that by the adjunction $\pi^{-1} \dashv \pi_*$, to give a morphism $\mathcal{O}_Y \rightarrow \pi_*\mathcal{O}_X$ is to give a morphism $\pi^{-1}\mathcal{O}_Y \rightarrow \mathcal{O}_X$. In which case the composition of $\varphi: \pi^{-1}\mathcal{O}_Y \rightarrow \mathcal{O}_X$ and $\psi: \tau^{-1}\mathcal{O}_Z \rightarrow \mathcal{O}_Y$ is

$$(\tau, \psi) \circ (\pi, \varphi): (\tau \circ \pi)^{-1}\mathcal{O}_Z \xrightarrow{\pi^{-1}\psi} \pi^{-1}\mathcal{O}_Y \xrightarrow{\varphi} \mathcal{O}_X.$$

Definition 2.3.2 (Schemes)

A **ringed space** is a Ring-space; we write RSp for $\text{Space}_{\text{Ring}}$.

An **affine scheme** is a ringed space which is isomorphic to $(\text{Spec } A, \mathcal{O}_{\text{Spec } A})$ for some ring A . A **scheme** is a ringed space $(X, \mathcal{O}_X)$ which is locally affine: for every $p \in X$ there is an open neighborhood U such that $(U, \mathcal{O}_X|_U)$ is affine. The topology of a scheme is called the **Zariski topology**.

Schemes form a full subcategory of RSp denoted Sch .

We can define the **global sections functor**

$$\Gamma: \text{Space}_{\mathcal{C}}^{\text{op}} \rightarrow \mathcal{C}, \quad \Gamma(X, \mathcal{O}_X) = \mathcal{O}_X(X).$$

A morphism $(\pi, \varphi): (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ determines a map $\varphi: \mathcal{O}_Y \rightarrow \pi_*\mathcal{O}_X$, which has component $\varphi_Y: \mathcal{O}_Y(Y) \rightarrow \mathcal{O}_X(\pi^{-1}Y) = \mathcal{O}_X(X)$. Thus we define $\Gamma(\pi, \varphi) = \varphi_Y$, turning Γ into a functor.

If X is a topological space and $U \subseteq X$ an open set, we may write $\Gamma(U, \mathcal{O}_X)$ instead of $\Gamma(U, \mathcal{O}_X|_U)$. In general we may write $(U, \mathcal{O}_X)$ instead of $(U, \mathcal{O}_X|_U)$.

Note:

Note that the global sections functor on sheaves is covariant, while the global sections functor on $\mathcal{C}$ -spaces is contravariant.

Lemma 2.3.3

There is a natural isomorphism

$$(D(f), \mathcal{O}_{\text{Spec } A}) \cong (\text{Spec } A_f, \mathcal{O}_{\text{Spec } A_f}).$$

Proof

Consider the canonical map $\iota: A \rightarrow A_f$, which induces the continuous injective $\iota: \text{Spec}(A_f) \rightarrow \text{Spec}(A)$. Its image consists of all prime ideals not intersecting $\{f^k\}_{k \geq 0}$, equivalently not containing f , so $D(f)$. It is also an open map: $D_{A_f}(g/f^n)$ contains all prime ideals of A_f not containing g/f^n ; since f^n is a unit, this is just all prime ideals not containing g . Thus $\iota D_{A_f}(g/f^n) = D(g) \cap D(f) = D(fg)$ which is open. Thus ι forms a homeomorphism $\text{Spec}(A_f) \xrightarrow{\sim} D(f)$.

Now we need to show that there is an isomorphism $\mathcal{O}_{\text{Spec } A}|_{D(f)} \cong \iota_* \mathcal{O}_{\text{Spec } A_f}$. Note that for $D(g) \subseteq D(f)$ we have

$$\mathcal{O}_{\text{Spec } A}(D(g)) = A_g, \quad \iota_* \mathcal{O}_{\text{Spec } A_f}(D(g)) = \mathcal{O}_{\text{Spec } A_f}(\iota^{-1} D(g)) = \mathcal{O}_{\text{Spec } A_f}(D(g)) = (A_f)_g \cong A_g.$$

So there is indeed an isomorphism, as desired. $\blacksquare$

Corollary 2.3.4

If $(X, \mathcal{O}_X)$ is a scheme and $U \subseteq X$ is open, then $(U, \mathcal{O}_X|_U)$ is a scheme.

Proof

For an affine scheme this is due to the previous lemma: because the $D(f)$ form a base, for any $p \in U$ there is an $f \in A$ such that $p \in D(f) \subseteq U$. Then $(D(f), \mathcal{O}_X|_{D(f)}) \subseteq (U, \mathcal{O}_X|_U)$ is an open affine subscheme.

In general let $p \in U$, then take $p \in V \subseteq X$ an affine neighborhood. Then $p \in U \cap V \subseteq U$ is an open subset of an affine scheme and thus there is an affine neighborhood $p \in W \subseteq U \cap V$. $\blacksquare$

Definition 2.3.5

If $(X, \mathcal{O}_X)$ is a scheme and $U \subseteq X$ open, then $(U, \mathcal{O}_X|_U)$ is called an **open subscheme** of X . If U is also an affine scheme, then U is an **affine open subscheme** or an **affine open subset**.

It is clear that for a scheme X , its affine open subsets form a base for the topology.

Definition 2.3.6

Let $\mathcal{C}$ be a category, a subcategory $\mathcal{C}_0 \subseteq \mathcal{C}$ is said to be **replete** if it is closed under isomorphisms: if $c \in \mathcal{C}_0$ and $f: c \xrightarrow{\sim} d$ is an isomorphism then both d and f are in $\mathcal{C}_0$.

Let $\mathcal{C}_0 \subseteq \mathcal{C}$ be a replete category. A $(\mathcal{C}, \mathcal{C}_0)$ -space is a $\mathcal{C}$ -space $(X, \mathcal{O}_X)$ such that for every $p \in X$, the stalk $\mathcal{O}_{X,p}$ is in $\mathcal{C}_0$. A morphism of $(\mathcal{C}, \mathcal{C}_0)$ -spaces is a $\mathcal{C}$ -space morphism $(\pi, \varphi): (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ such that for

every $p \in X$ the morphism of stalks $\varphi_p: (\pi^{-1}\mathcal{O}_Y)_p \cong \mathcal{O}_{Y,\pi(p)} \rightarrow \mathcal{O}_{X,p}$ is in $\mathcal{C}_0$.

Note:

We can write $\varphi_p: \mathcal{O}_{Y,\pi p} \rightarrow \mathcal{O}_{X,p}$ more explicitly. For $\pi(p) \in V \subseteq Y$ open we have a map

$$\varphi_V: \mathcal{O}_Y(V) \rightarrow \mathcal{O}_X(\pi^{-1}V) \rightarrow \operatorname{colim}_{p \in U} \mathcal{O}_X(U) = \mathcal{O}_{X,p}.$$

These then assemble into a cone $\mathcal{O}_Y(\bullet) \Rightarrow \mathcal{O}_{X,p}$ which then limits to $\mathcal{O}_{Y,\pi(p)} \rightarrow \mathcal{O}_{X,p}$.

Definition 2.3.7

Define **LocRing** to be the category of local rings $(A, \mathfrak{m})$. A morphism $\varphi: (A, \mathfrak{m}) \rightarrow (B, \mathfrak{n})$ is a ring morphism $A \rightarrow B$ such that $\varphi(\mathfrak{m}) \subseteq \mathfrak{n}$.

A **locally ringed space** is a $(\mathbf{Ring}, \mathbf{LocRing})$ -space. We denote the category of locally ringed spaces by **LocRSp**.

Note:

Importantly, a locally ringed space is not a **LocRing**-space. In fact, generally a locally ringed space's sections will not be local rings.